

A MODEL OF OPTIMAL SCIENCE FUNDING: TARGETING THE CAPABILITY-RESOURCE GAP

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ABSTRACT. The grant funder’s challenge is to maximize the impact of a fixed budget under uncertainty about the researchers who might receive it. We develop a Bayesian model of this challenge: researchers differ in their capabilities and in their resources; research output requires both, and funding compounds capability across rounds. We show that the output-maximizing allocation of a round’s budget funds the capability-resource gap: grants increase with capability and decrease with existing resources. Track records underdetermine the gap, because output confounds the two; what a funder needs is a signal of capability separate from resources, and peer review is one source of such a signal. We find that the value of that signal increases with how unequally the gaps are distributed, which capability inequality mainly drives, and with the signal’s informativeness, and that under heavy-tailed capability distributions even a noisy signal captures most of that value. Uniform seed grants and lotteries, which forgo targeting, lose little research output exactly where review is worth little, and most of what a partial lottery loses comes from discarding the review ranking rather than from chance. Two features of a field, the funder’s budget relative to researchers’ resources and the inequality of the gaps, locate a funder on a map of these results.

Significance. Research funders divide fixed budgets among researchers whose abilities and needs they cannot observe. Whether to concentrate funds or spread them, whether peer review is worth its cost, and whether lotteries should replace it are unsettled. We model funding as a sequential decision under uncertainty and show that the output-maximizing allocation funds the gap between each researcher’s capability and their resources, that track records alone underdetermine that gap, and that what peer review is worth and how lotteries and seed grants perform turn on two features of a field: the funder’s budget relative to researchers’ resources, and how unequal the gaps are, which mostly reflects how unequal capability is. We show where representative funders fall on these two features.

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The United States spent close to a trillion dollars on research and development in 2024, roughly a fifth of it funded by the federal government (National Center for Science and Engineering Statistics 2026; National Science Board 2026), and how best to allocate that money remains unsettled. Whether funding should be concentrated among a few researchers or spread across many is a long-running dispute (Aagaard et al. 2020). How well peer review predicts research outcomes is contested: among funded NIH grants, percentile scores predicted subsequent productivity barely better than chance (Fang et al. 2016), while studies with other designs find modest but real predictive power (Li and Agha 2015; Park et al. 2015). Several funders now allocate part of their budgets by lottery, in forms that range from a tie-break among top-ranked proposals to a draw among all applicants above a threshold (Liu et al. 2020; Heyard et al. 2022; Feliciani et al. 2024). And a randomized trial of winning a grant found no clear effect on the recipients’ publications over the following four years (Barnett et al. 2024), which raises the prior question of when a marginal grant adds output at all.

These disputes concern one problem: what a funder with a fixed budget and imperfect knowledge of researchers should do. The empirical studies each measure one input to that decision, how well scores predict outcomes among funded grants or how research output responds to grant size, and none can say what the funder should do with what it knows; that depends on what the funder is uncertain about, what its information is worth, and how the answer changes with the budget. We treat the funder’s problem as a sequential decision under uncertainty and analyze it using a Bayesian approach. Researchers differ in capability, what they know and can do, and in resources, what they have to work with; output requires both and is limited by the scarcer of the two; the funder observes output, and the noisy judgments of reviewers, but neither attribute directly. A grant adds resources in the round it is given and, because doing research builds expertise, raises capability in later rounds. Each round, the funder observes the output that results, updates its beliefs about the researchers, and allocates again.

Three results follow. First, the allocation of a round’s budget that maximizes expected research output funds the gap between each researcher’s capability and their resources: grants increase with capability and decrease with existing resources, and the budget sets how many researchers are funded at all. Two heuristics a funder might use, funding the track record and funding the under-resourced, each track one of the two quantities, and each can perform worse than simply dividing the budget uniformly across all researchers. Second, track records under-determine the gap, because output confounds capability with resources; what the funder needs is a signal of capability separate from resources, and peer review is

one source of such a signal. The value of that signal, and how seed grants and lotteries perform relative to targeted funding, depend on the same two features of a field: how much of researchers' resources the funder's budget can supply, and how unequally the capability-resource gaps are distributed, which in every setting we ran is mostly a matter of how unequal capability is. Where the budget is tight and capability heavy-tailed, targeting is worth a great deal, even noisy review captures most of that value, and lotteries give up much of it; where the budget is ample and capability evenly spread, little research output is at stake and spreading funds widely loses almost nothing. Within the model, no funding mechanism is best in every field. Third, the two features can be estimated from what funders already know, and we place representative funding programs on the map they define.

Two assumptions bound these results: grants are used up in the round they are given, with whatever lasts from them entering as capability, and capability and resources are complements. The objective is aggregate expected research output; diversity of directions, risk, and fairness are outside it, and "optimal" throughout means output-maximizing under that objective. The model is deliberately simple. Its contribution is to the theoretical understanding of the funder's problem: it shows that the disputes turn on two features of a field, and it says which way each dispute goes in each kind of field.

RESULTS

The optimal allocation funds the capability-resource gap. Consider first a funder that knows every researcher's capability K_i and resources R_i and divides one round's budget B so as to maximize that round's expected output, which for researcher i is $AK_i R_i / (K_i + R_i)$, where A is a productivity constant that sets the units of output and R_i is the sum of baseline resources R_{i0} and the grant g_i (Materials and Methods). The marginal value of a dollar to researcher i is $AK_i^2 / (K_i + R_{i0} + g_i)^2$: it increases with their capability and decreases with the resources they already hold, grants included. An allocation is optimal exactly when no dollar can be moved to a researcher who would produce more with it, and solving that condition gives the gap rule (SI Text 1, Proposition 1):

$$g_i^* = \max(c K_i - R_{i0}, 0),$$

where $c > 0$ is the value at which the grants sum to the budget. Each researcher's target is c times their capability; the rule fills each positive gap between target and resources and grants nothing to the rest. The rule does not depend on the harmonic form. For any production function with constant returns to scale and diminishing returns to resources, $F(K, R) = AK h(R/K)$ with h increasing and

strictly concave, equal marginal value among the funded means a common ratio of resources to capability, and the same rule follows with c set by the budget (SI Text 1, Proposition 1); the harmonic mean is the instance we simulate. The condition $R_{i0} < cK_i$ draws a frontier through the population, separating the funded from the unfunded (Fig. 1). When the budget is small, c is small and few researchers are funded; a larger budget raises c , moves the frontier outward, and funds more researchers with larger grants, until at a large enough budget funding extends across nearly the whole population.

The gap rule explains why each heuristic fails. Funding the track record fails because output reflects resources as well as capability: a productive, well-resourced researcher has a small or negative gap, and an additional dollar adds little to their output. Funding the under-resourced fails for the opposite reason: scarcity is no evidence of capability, and dollars given to researchers of low capability add little output however scarce their resources. Each heuristic can produce less output than dividing the budget equally (SI Text 1, Examples 1 and 2; Fig. S1 shows both schemes on the population of Fig. 1).

How much targeting matters depends on the budget. Expected output increases with resources at a diminishing rate and never exceeds AK_i , so when the budget is large enough to bring every researcher’s output near that bound, the optimal allocation and equal division differ little. The value of targeting, the expected output the optimal allocation adds over equal division, vanishes as the budget grows (SI Text 1, Corollary 2), and in every one of several hundred simulated populations it also fell with the budget throughout, measured as a share of uniform funding’s own gain (Fig. S2). Targeting matters where the budget is small relative to researchers’ resources, and this dependence on the budget recurs in every result below.

Proposition 1 concerns one round. Over several rounds a grant also raises the recipient’s future capability, and the allocation that maximizes output over the whole horizon need not take the gap form. Numerically it nearly does. Over two rounds with complete information, the exact dynamic optimum, free to choose its recipients and how much of the budget to spend in each round, exceeds the gap rule applied round by round with an even split by at most 1.4% of the research output that funding adds, across every capability tail, compounding rate, and budget we sweep, and its first-round grants correlate with the gap rule’s at 0.98 or above (SI Text 1, Table S4). We therefore use the round-by-round gap rule as the complete-information benchmark throughout, and “optimal allocation” below refers to it.

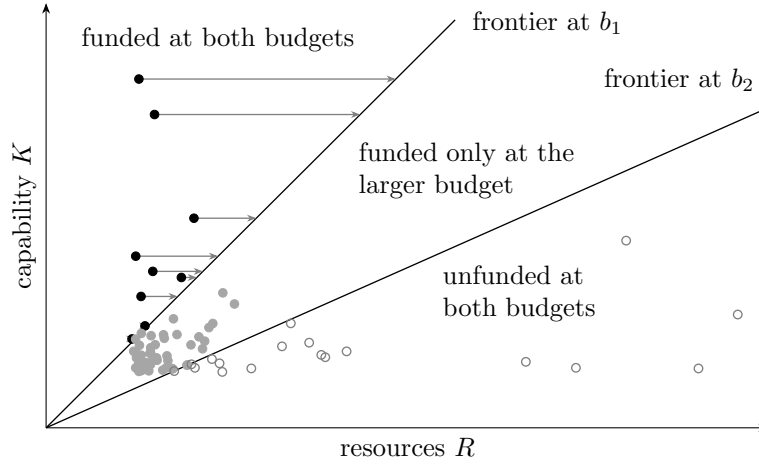


FIGURE 1. The gap rule in one population at two budgets. Each point is a researcher, placed by resources and capability. The two lines are the funding frontiers $R = cK$ at a smaller budget b_1 and a larger budget b_2 ; a larger budget raises c and flattens the frontier. Researchers to the left of a frontier are funded at that budget: black points at both budgets, gray points only at the larger budget, open points at neither. Each arrow is a grant at the smaller budget, which moves its researcher exactly to that budget's frontier; at the larger budget every black and gray researcher receives a grant that moves it to the outer frontier (arrows not drawn).

What the funder can learn: track records and peer review. A real funder observes research output, not capability or resources. Expected output is one number produced by the two jointly, so a researcher of high capability and modest resources and a researcher of modest capability and abundant resources can produce identical records, and these are exactly the researchers the gap rule treats most differently: the first may have the largest gap in the population, the second none. Even expected output gives only a lower bound on capability: since expected output never exceeds AK_i , a researcher whose expected output is λ_i has $K_i > \lambda_i/A$, and realized output, a Poisson draw around λ_i , bounds it less. The problem is worst where the gap is largest. When resources are small relative to capability, expected output is approximately AR_i , proportional to resources and nearly independent of capability (Fig. 2A), so on small grants output carries little information about capability, and only grants comparable to capability itself make records informative about it.

A Bayesian funder relying on records alone therefore allocates far from the gap rule, and further rounds of records improve its allocation only slowly: its expected output falls short of the complete-information optimum by 46% of what that optimum adds in the first round and still by 19% in round twenty, while a funder

that also holds a single review score for each researcher, observed once at the start, is 8% short in round one and 3% by round twenty (Fig. 2*B*); in the same runs, the correlation of the records-only funder’s grants with the optimal grants rises from 0.08 to 0.32, against 0.83 for the review funder in round one. The reason lies in the model’s structure. Capabilities compound while resources do not, so each researcher’s output approaches the level their resources sustain, and output at that level carries less and less information about capability.

What the funder lacks is a signal of capability separate from resources. Peer review can provide one: a noisy signal, but one whose noise does not come from resources. We model it as each researcher’s capability plus Gaussian noise, observed once (Materials and Methods). We measure the value of review as the additional expected output of a funder allocating with the signal over the same funder allocating without it, and we report it as the share of the records-only shortfall that the signal recovers, where the records-only shortfall is the difference between the records-only funder’s output and the output a funder with complete information would produce. That value depends on the field (Fig. 2*C*). Review is worth most where capability is heavy-tailed and the budget is tight, and under heavy-tailed capability even a signal of moderate informativeness captures most of its value: at the default signal noise, review retains 82% of its maximum value when the capability distribution’s tail parameter is $\alpha_K = 1.3$, 72% at the default distribution ($\alpha_K = 2$), and 45% where capability is spread more evenly ($\alpha_K = 3.5$); at three times the default noise, 52%, 33%, and 8%. The three distributions have the same mean capability, so the tail parameter varies how dispersed and how heavy-tailed capability is, not its level. The reason is twofold. Where capability is heavy-tailed, most of the output a funder can add comes from getting resources to the few researchers of unusually high capability, and precisely those researchers are the easiest to identify: they stand far apart from everyone else, so even a noisy signal separates them. Where capability is spread more evenly, no researcher matters much more than another, so review of any informativeness adds little; and where the budget is ample, information about whom to fund has little left to add. This ordering of review’s value by field holds across production functions from Cobb-Douglas to Leontief (SI Text 4), and it survives the checks a skeptic would ask for: it holds when review is compared across fields at equal rank correlation between score and capability rather than at equal noise, for applicant pools of 25 to 200, for heavier and lighter tails of the resource distribution, when the review score partly reflects resources rather than capability alone, when review noise is multiplicative rather than additive, when review is repeated afresh every round rather than observed once, and across resource signals from nearly exact to nearly uninformative (SI Text 5). Of these variants, multiplicative review noise, a

constant relative error, is the one that narrows the heavy-tailed field’s advantage, because a constant relative error is a large absolute error for the researchers who matter most there: at equal informativeness review recovers 0.66 of the shortfall in the heavy-tailed field against 0.49 in the evenly spread field, where the additive signal gives 0.87 against 0.55. What review supplies, in every one of these variants, is information about capability that output cannot supply on its own; peer review is one source of such information, and the results apply to any other, such as a pilot grant large enough to make output informative. One feature changes how much any of this is worth: the association between capability and resources. Where the capable already hold the resources, the gaps are small and even, and both targeting and review are worth less; where resources sit with the less capable, both are worth more. One condition qualifies it: the funder must weight review at its true informativeness. A funder that overtrusts review, treating a noisy signal as reliable, can do worse than one that ignores review altogether, and overtrusting review loses more output than undertrusting it (Fig. S3).

Spreading funds: seed grants, lotteries, and concentration. Two alternatives to targeting recur in funding policy, and each has a rationale. Uniform seed grants give every researcher the same amount, so that a researcher whom review misjudges still receives something; in practice a funder gives out part of its budget as seed grants and targets the rest. Lotteries give grants to applicants chosen at random, usually from among those who pass a review threshold; their advocates argue that review cannot reliably rank the applicants it deems fundable, that review is costly, and that review is biased (Fang and Casadevall 2016; Roumbanis 2019). Both alternatives give up part of targeting on purpose, so both give up part of the value of targeting, and that value depends on the field. For a funder allocating with records and review, targeting adds more research output than uniform funding’s entire gain over no funding in a heavy-tailed field with reliable review and a tight budget, and about a quarter of that gain in the default field at the largest budget we sweep (SI Text 3). What the alternatives save, in the cost of review and in the effort researchers spend on proposals, lies outside our model (Gross and Bergstrom 2019).

Consider seed grants first. The output lost increases faster than the fraction of the budget given out as seed grants: giving out half of the budget as seed grants loses more than twice the output that giving out a quarter loses (Fig. 3A). The size of the loss is set by what targeting is worth in the field: when a review-informed funder gives out three quarters of its budget as seed grants, it forfeits about 11% of the research output its funding adds in a heavy-tailed field with reliable review (17% at the smallest budget we sweep), and about 3.5% in the default field.

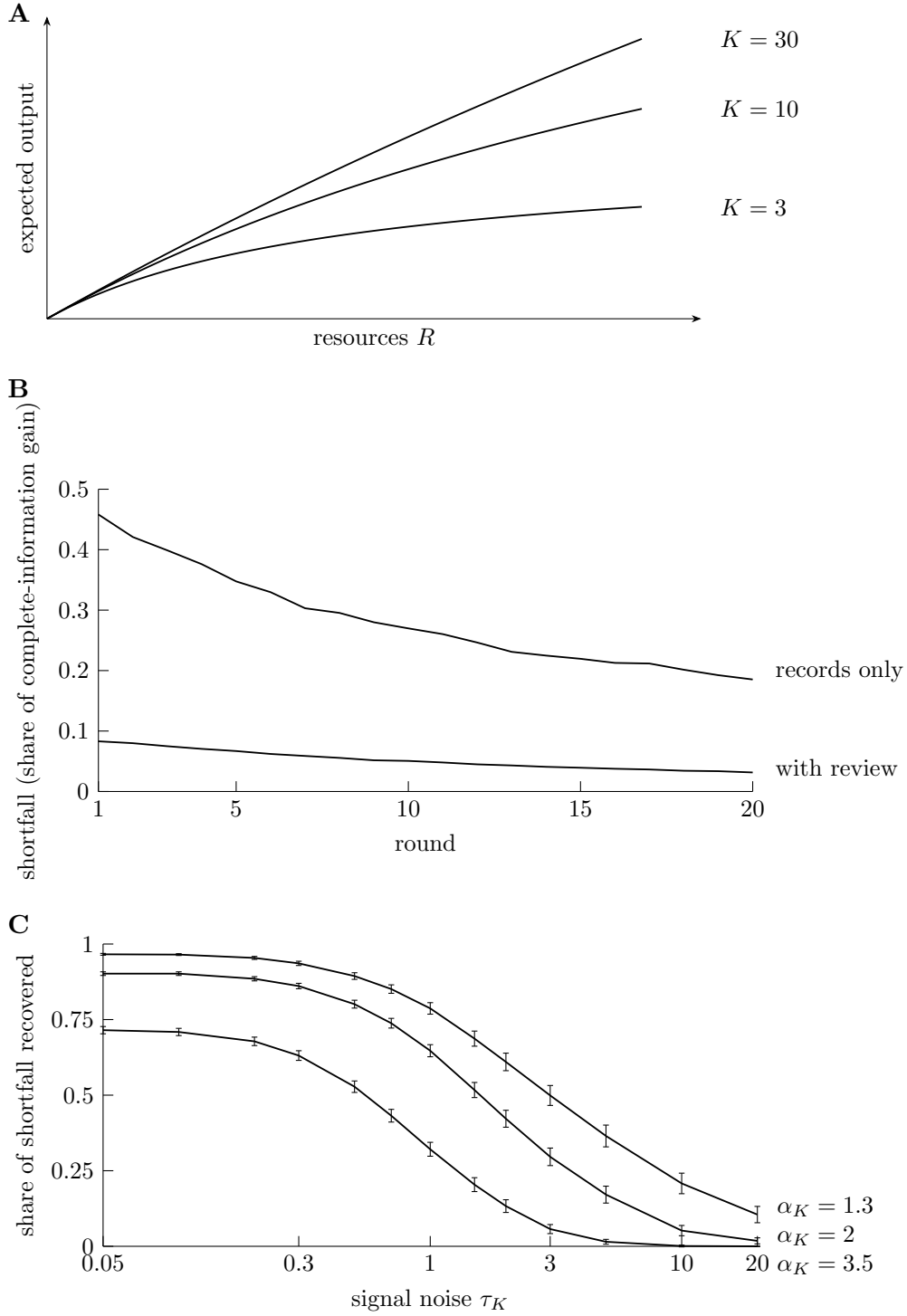


FIGURE 2. What the funder can learn. (A) On small grants, research output carries little information about capability: expected output against resources for researchers of capability $K = 3, 10$, and 30 ; the curves coincide where resources are small relative to capability. (B) Records do not catch up: the shortfall of each funder's expected output below the complete-information optimum, as a share of what complete-information funding adds, round by round in the default field; the records-only funder's shortfall falls from 46% in round one to 19% by round twenty, while the funder with a single review score observed at the start is at 8% in round one and 3% by round twenty. (C) Review is worth most where

Now consider lotteries. The lotteries funders run are partial: review sets a fundable pool, and chance picks the grantees within it. Such a lottery gives up two things, the review ranking within the pool and the certainty of who is funded, and the question is what each costs. We evaluate five schemes in a single round (Fig. 3*B*): equal division among the fifth of researchers with the highest review scores (selection by review); equal division among the top two fifths; the partial lottery that some funders run, which funds half of the top two fifths at random with full-size grants; uniform funding; and a lottery over all researchers with full-size grants. In a heavy-tailed field with a tight budget, selection by review gains about three quarters of what the optimal allocation gains when review is reliable and about half when review is very noisy. The partial lottery gains less, and most of the difference comes from ignoring the review ranking within the top two fifths rather than from the randomness itself: at the default noise, the partial lottery falls 0.18 short of selection by review, of which ignoring the ranking accounts for 0.14 and the randomness for 0.04. The lottery over all researchers gains least, less than uniform funding. In an evenly spread field with a large budget, the ordering reverses: uniform funding gains more than four fifths of what the optimal allocation gains, and every scheme that concentrates grants, whether by review score or by chance, gains less. A lottery among applicants above a review threshold therefore loses little output only where the review ranking it ignores is worth little: replacing that ranking by chance loses a few hundredths of the optimal allocation’s gain in an evenly spread field or when review is very noisy, and about a sixth of that gain in a heavy-tailed field with reliable review. In an evenly spread field, the better alternative to review is small grants to many researchers; a lottery among a few applicants above a review threshold produces less.

The randomness itself costs output for a general reason. A lottery gives each applicant in its pool a chance at a full grant instead of a certain smaller one, and on average the applicant receives the same amount either way; because expected output increases with resources at a diminishing rate, the certain grant produces more. Dividing a pool’s money equally among the pool therefore produces more expected output than any lottery over the same pool, provided that grants are divisible and that output rises with resources at a diminishing rate at every grant size (SI Text 2, Proposition 2). Where a project needs a minimum viable grant, the first condition fails and a lottery over full awards can beat dividing the same money into grants too small to use (SI Text 2). The cost of the randomness is small in our simulations; the cost of the discarded ranking is the one that depends on the field.

Seed grants and lotteries both spread funds more widely than targeting does, and both raise the older question of whether funding should be concentrated among a few researchers or spread across many. The production function does not settle that question. Stronger complementarity between capability and resources concentrates the review-informed funder’s grants when the budget is tight in a heavy-tailed field, spreads them when the budget is ample, and has a non-monotone effect where capability is evenly spread (Fig. S5; SI Text 3). How widely funds should be spread is set jointly by the production function, the budget, and the field’s capability distribution.

DISCUSSION

Our results isolate one quantity as key to optimal grant funding: the value of targeting researchers with the greatest gap between their capabilities and their resources. Two features of a field determine this value. The first is the funder’s budget relative to researchers’ own resources: where the budget is small, dollars are scarce and where they go matters; where the budget can relieve most researchers’ bottlenecks, the optimal allocation and equal division produce nearly the same research output. The second is how unequally the capability-resource gaps are distributed. Across every setting we ran, the value of targeting is close to a function of the inequality of the optimal grants alone (SI Text 5), and what makes the gaps unequal is mainly how unequal capability is: where a few researchers can produce far more than the rest, getting resources to those few is most of what a funder can add. The association between capability and resources works through the same channel: where the capable already hold the resources, the gaps are small and even, and targeting is worth less. Every result above is a statement about this quantity: what records and review contribute to capturing it, and what seed grants and lotteries give up of it. Two results lie outside it. How widely a review-informed funder should spread its grants depends, in addition, on how strongly capability and resources complement each other (SI Text 3, Fig. S5); and the timing of spending over the horizon, which we analyze in SI Text 6, adds little research output beside the choice of recipients in every setting we test, subject to the limits stated there.

These results have a practical consequence for funders. Within the model, no funding mechanism is best in every field. Peer review, seed grants, and lotteries each perform better in some fields than in others, and the two features of the field decide which, so a funder choosing among mechanisms should begin by locating its field on the two (Fig. 4). The first feature is observable. A funder’s budget relative to the field’s own resources can be read from administrative data: its annual awards to the field against the field’s research spending from every other source.

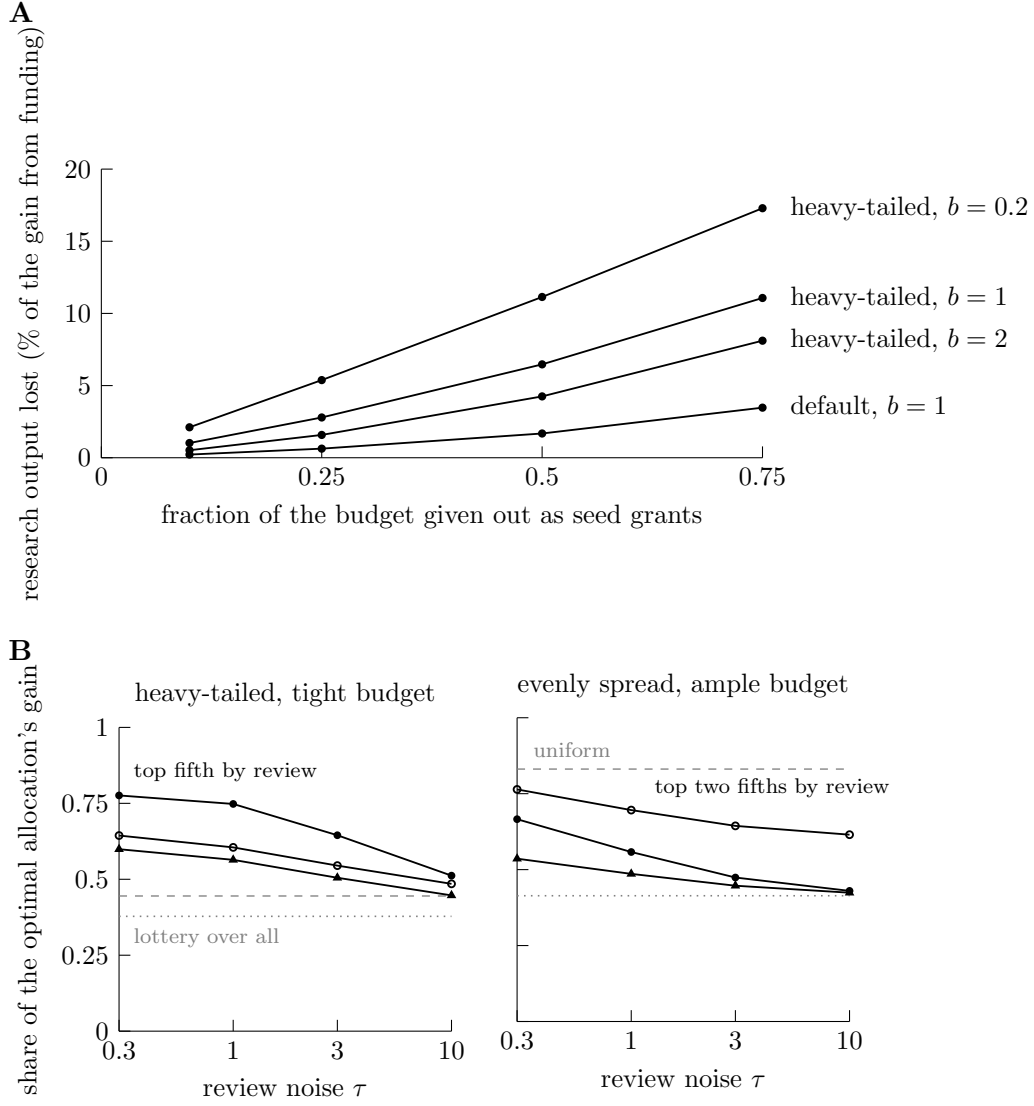


FIGURE 3. Spreading funds. (A) The research output lost to uniform seed grants, as a percent of the research output that the funder's grants add over no funding, when a review-informed funder gives out a fraction of its budget as equal grants to all researchers and targets the rest; the loss grows faster than the fraction, and its size is set by what targeting is worth in the field. (B) What five funding schemes gain, by field: each scheme's gain over no funding as a share of the optimal allocation's gain, as review noise increases, in a heavy-tailed field with a tight budget (left) and an evenly spread field with a large budget (right). Filled circles: equal division among the top fifth by review score; open circles: among the top two fifths; triangles: the partial lottery; dashed: uniform funding; dotted: a lottery over all researchers. Settings: SI Materials and Methods.

The second feature is not. The inequality of the gaps depends on capabilities the funder cannot see and on the budget itself, and the inequality of the optimal grants, the quantity that organizes our results (SI Text 5), is not one a funder can compute. Three observable quantities stand in for it. The inequality of research output in the field is the natural proxy for the inequality of capability, though where grants are small output tracks resources rather than capability, so this proxy is imperfect in the direction of the resource distribution. The dispersion of review scores over the whole applicant pool is a second, with the same caveat about noise. The most direct is the response of output to grants large enough to relieve the bottleneck: on a grant comparable to the recipient's capability, output becomes informative about capability (Fig. 2A), so a program of substantial pilot grants measures the gaps, whereas small seed grants do not, because on small grants output tracks resources for every recipient. How existing funding is distributed completes the estimate: where resources already sit with the capable, the gaps are smaller and more even than capability inequality alone would imply.

Fig. 4 places four kinds of funding program on the budget axis by order of magnitude, from 2023 and 2024 budget figures (Materials and Methods). A national biomedical agency that supplies a large share of its field's research support sits near a budget equal to the field's other resources; a national science agency that funds its fields more thinly sits near a tenth of that; a continental excellence council near a few hundredths; and a private foundation funding one field among many at a hundredth or below. Where a field lies on the inequality axis is the estimate each funder must make, and the range of tail parameters estimated for research productivity spans most of the axis, so the placements say which regime a funder's budget puts it in, not where it sits within the regime. The regimes differ sharply. A foundation or council with a small budget in a heavy-tailed field faces the largest value of targeting per dollar: review directed at identifying the few researchers of unusually high capability is worth most, even noisy review recovers most of that value, seed grants give up the most output, and a lottery among applicants above a review threshold gives up about a sixth of what selection by review gains. An agency whose budget approaches the field's other resources, in a field where capability is evenly spread, faces a small value of targeting: review of any informativeness adds little, seed grants and lotteries give up little, and the model's best use of the budget is small grants to many researchers.

Three further implications concern peer review. Since most of review's value comes from separating the few researchers of unusually high capability from the rest, and fine distinctions among the many applications of comparable merit add little research output, a review system that spends most of its effort ranking the middle

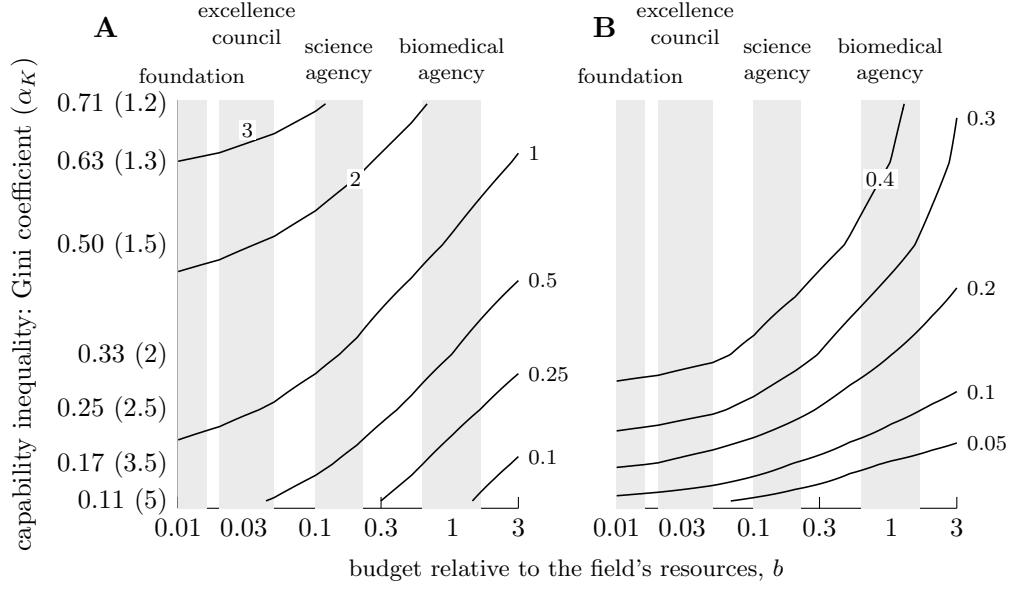


FIGURE 4. Where a funder stands. (A) The value of targeting: the complete-information optimum's gain over uniform funding, as a share of uniform funding's gain over no funding. (B) The value of review at the default noise, as a share of the research output that review-informed funding adds. Both by the funder's budget relative to the field's resources and the inequality of capability, at fixed mean capability; contours are labeled by value. Shaded bands place four kinds of program on the budget axis by order of magnitude, from 2023 and 2024 budget figures (SI Materials and Methods); where a field lies on the vertical axis is the estimate each funder must make, and the tail parameters estimated for research productivity lie in its upper part. Two rounds, $\tau_K = 1$, 50 populations per cell; contours interpolated between the 63 cells. Settings: SI Materials and Methods.

of the distribution spends that effort where the model locates the least output. Second, the finding that percentile scores barely predict productivity comes from funded NIH grants with scores in the top fifth (Fang et al. 2016), and the studies that find modest predictive power also use funded grants (Li and Agha 2015); both measure prediction among applicants that review has already selected, within the range where the model expects review to add least, so a weak association there is consistent with review of considerable value over the whole applicant pool. Third, a funder that treats review as more reliable than it is loses more output than a funder that treats review as noisier than it is, so a funder uncertain about review's accuracy should err toward undertrust.

The model holds several things fixed, and its results hold within those limits. There is one funder, and researchers do not respond strategically, so competition among funders, proposal effort, risk choice, and the cost of review are outside the model; the contest models of grant competition cover that ground (Gross and Bergstrom 2019, 2025), and the mechanism-design treatment of grants by Carnehl et al. (2024), in which the funder must also elicit information from applicants who bear the cost of applying, is the natural complement to a model in which the funder’s information arrives for free. Sorzano and Pueche-Granados (2026) also treat funding as a decision under heavy-tailed returns and find more room for lotteries than we do; their heavy tail is in observed impact, whereas ours decomposes output into latent capability and resources, and that decomposition is what gives review its value here. Output is a single number per round, so projects do not differ in kind and exploring new directions has no value in the model (Avin 2019); nor do a researcher’s proposals differ in quality, so this is a model of investigator-level funding, and project-specific quality that review might observe is the natural extension. Grants are divisible, so a minimum viable grant size, which can favor a lottery over full awards, is outside the model (SI Text 2). The review signal is noisy but unbiased; bias and conservatism in review, which are among the strongest arguments for lotteries, are not modeled. Capability is a state, not a trait: it holds whatever a researcher has accumulated that raises output, expertise, methods, a trained group, and doing research adds to it; a funder that reads it as innate ability misreads the model. The asymmetry between the two inputs is an assumption with consequences: capability persists and grows, resources recur and do not persist, so a grant’s lasting effect enters only through capability. Durable equipment, data, and infrastructure violate this asymmetry. The one-round results do not depend on it; the results about compounding and about the timing of spending (SI Text 6) do. The compounding here is of capability through doing research, and differs from the cumulative advantage in which winning a grant raises the chance of winning the next (Bol et al. 2018); the two would need to be separated in any empirical test. Inputs are complements, and the timing results depend on complementarity while the information results do not. The numbers we report are comparisons across the parameter values we sweep, and none is calibrated to a real funder; what applies to a real field is the qualitative features of the dynamics that organize the results.

The model makes four predictions that data on funded research can test. The effect of winning a grant on output should be larger for recipients whose other resources are small relative to their capability, so a randomized trial of funding, such as the one that found no clear average effect (Barnett et al. 2024), should find its effect concentrated among under-resourced recipients, and a null average is what the

model predicts where most recipients are already well resourced. Measures that stand in for capability, such as prior record or review score, should predict research output better the larger a grant is relative to the recipient’s other resources, because on small grants output is nearly independent of capability. How well review scores predict output, measured over the whole applicant pool rather than among funded grants, should be higher in fields whose output distributions have heavier tails. And where funders run lotteries beside review, the output lost to the lottery should be larger in heavier-tailed fields. Each of the remaining predictions can be tested with data that funders already hold: the first by comparing how well review scores predict output across grant sizes within one applicant pool; the second by comparing that predictive power across fields, scored over the whole applicant pool where the outcomes of funded and unfunded applicants can both be observed; the third by comparing the outputs of lottery-funded and review-funded grants where a funder runs both.

A funder with a fixed budget and imperfect knowledge of researchers faces one problem, and our model gives it one answer. The output-maximizing allocation funds the gap between each researcher’s capability and their resources. Records alone underdetermine that gap, because output does not identify which input binds; a signal of capability separate from resources does, and peer review is one source of it. What that signal is worth, and how seed grants and lotteries perform relative to targeted funding, turn on the same two features of a field, and the map of those features has regimes in which the answers differ. A funder deciding among review, lotteries, and seed grants should therefore begin by asking which kind of field it is funding.

MATERIALS AND METHODS

Model. A funder faces n researchers over T funding rounds. Researcher i has capability K_i and resources R_i ; initial values are drawn from Pareto distributions with tail parameters α_K and α_R (smaller values give more unequal fields; below 2 the variance is infinite), coupled through a Gaussian copula with parameter ρ (a Pearson correlation between Pareto variables is undefined when their variances are infinite; $\rho = 0.5$ and 0.8 correspond to rank correlations of 0.48 and 0.79). Capability is a state variable: it holds whatever a researcher has accumulated that raises output, and doing research adds to it. In each round, researcher i produces research output drawn from a Poisson distribution with mean $\lambda_i = AK_iR_i/(K_i + R_i)$, proportional to the harmonic mean of the two inputs, where the productivity constant A sets the units of output (the simulations set $A = 1$); output increases in both inputs and is limited by the scarcer. Between rounds capability compounds, $K_i \leftarrow K_i + \epsilon \lambda_i/A$, at rate ϵ . Resources do not accumulate:

$R_i = R_{i0} + g_i$, where the baseline R_{i0} recurs every round and the grant g_i is consumed in the round it is given. The unit of analysis is the researcher, not a researcher-project pair. The funder’s total budget over the horizon is $b n \mathbb{E}[R_0]$, so at budget scale $b = 1$ it equals the community’s expected baseline resources for one round; the budget does not depend on the horizon.

Information and strategies. The funder knows the distributions but observes no researcher’s capability or resources. It observes each researcher’s realized output, a Poisson draw with mean λ_i , as it accumulates; a resource signal $R_{i0} + \eta_i$, observed once, with η_i Gaussian with standard deviation τ_R ; and a review signal $K_i + \xi_i$, observed once at the start by the strategies that use it, with ξ_i Gaussian with standard deviation τ_K and independent across researchers; a reviewer’s misjudgment of a researcher therefore persists, and a variant with a fresh review every round is in SI Text 5. Bayesian strategies hold a posterior over each researcher’s capability and baseline resources, computed from these likelihoods by importance sampling with M particles drawn from the prior ($M = 1,000$ for Fig. 2B and C, Fig. 4, and the sensitivity analysis, and 200 elsewhere; the convergence of the results in M is checked in SI Materials and Methods), update it after each round, and allocate each dollar to the researcher whose expected output, given the posterior, rises most with it. They differ in information (with or without the review signal) and in planning. The myopic funder maximizes the current round’s expected output. The forward-looking funder is a certainty-equivalent receding-horizon planner: each round it chooses the current allocation together with a plan for the remaining budget so as to maximize expected output over the remaining horizon, treating future output as equal to its expectation, and it re-plans after observing the round’s output; it does not value the information that its own grants will generate (SI Text 6). The complete-information benchmark applies the gap rule to each round’s share of the budget with the true capabilities and resources. Three baselines complete the set: no funding, against which every result is normalized; uniform funding; and track-record funding, which allocates in proportion to the previous round’s output. Seed grants and lotteries modify these strategies as described in Results (SI Materials and Methods, Table S1).

Parameters and scoring. Tail parameters range from 1.3 to 3.5, a range that includes the Pareto tails near 1 to 2 estimated for research productivity (Lotka 1926; Nicholls 1989); since productivity is an imperfect guide to capability (Results), the range extends well beyond those estimates; the budget scale b from 0.01 to 3, with larger budgets in the timing analyses (SI Text 6); the compounding rate ϵ from near zero to 0.85; review noise τ_K from 0.05 to 20; the horizon from one round to ten. Default values are $T = 2$, $n = 50$, $\epsilon = 0.1$, $b = 1$, $\alpha_K = \alpha_R = 2$,

$\tau_K = \tau_R = 1$, $\rho = 0$. Each strategy is scored by the sum of expected outputs along its run, which equals realized output averaged over repetitions of the same allocations, and strategies are compared on identical simulated populations with identical signal and output draws, so comparisons are paired. Results are reported as shares of a named comparison, most often the research output that funding adds: a strategy's expected output minus the expected output under no funding. Analytic results (the gap rule, its corollaries, and the lottery proposition) are proved in SI Text 1 and SI Text 2; the settings behind each figure are in SI Materials and Methods. Code and data are available in the project repository.

Placing programs on the budget axis. The budget scale b of a real program is its annual awards to a field divided by the field's annual research spending from every other source, which stands in for the baseline resources of the field's researchers; the estimate counts money and not time, and treats each funder's field as the disciplines it mainly funds, so it is an order of magnitude. United States universities spent \$108.8 billion on research and development in fiscal year 2023, \$55.2 billion of it in the life sciences, and received \$33.1 billion from the Department of Health and Human Services, chiefly the National Institutes of Health, and \$6.7 billion from the National Science Foundation (National Center for Science and Engineering Statistics 2025): a biomedical agency of that size has b near 1 (\$33 billion against \$22 billion from other sources), and a science agency of that size funding the other half of academic research has b near 0.15. The European Research Council's budget of 16 billion euros over 2021 to 2027 (European Research Council 2025), about 2.3 billion a year, against 86 billion euros of higher-education research spending in the European Union in 2024 (Eurostat 2025), gives b near 0.03. A private foundation awarding on the order of \$100 million a year across several fields has b at or below 0.01 in any one of them.

SUPPORTING INFORMATION APPENDIX

The SI Appendix holds what the main text cites without proof or states in one sentence: the model details, strategies, parameter ranges, the particle audit, and the settings behind every figure (SI Materials and Methods); the derivation of the gap rule and its comparison with the sequential optimum (SI Text 1); the lottery proposition (SI Text 2); four extended results with Figs. S2 to S5 (SI Text 3); the robustness of the results to the production function (SI Text 4); the sensitivity of the value of targeting and of review to the parameters the main sweeps hold fixed and to the signal model (SI Text 5); and the timing of funding (SI Text 6). The table below maps each main-text statement to the section that supports it.

Main-text statement	Where in the SI
Model, signals, strategies (Table S1), parameter ranges, copula, particles (Table S2), scoring, settings behind every figure (Table S3), code	SI Materials and Methods
The output-maximizing allocation of a round's budget funds the gap: $g_i^* = \max(cK_i - R_{i0}, 0)$	SI Text 1, Proposition 1
The funded set and every grant increase with the budget; the funding frontier	SI Text 1, Corollary 1
The value of targeting decreases in the budget and vanishes as the budget grows	SI Text 1, Corollary 2; SI Text 3, Fig. S2
Each heuristic can perform worse than uniform division	SI Text 1, Examples 1 and 2, Fig. S1
The round-by-round gap rule is within about one percent of the sequential optimum	SI Text 1, Table S4
Equal division outperforms any lottery over the same pool	SI Text 2, Proposition 2
Targeting's value relative to uniform funding, across budgets and fields	SI Text 3
Overtrusting review loses more output than undertrusting it	SI Text 3, Fig. S3
Large grants reveal capability; the source of the shift of spending toward later rounds	SI Text 3, Fig. S4
The production function does not settle concentration	SI Text 3, Fig. S5
The ordering of review's value by field holds from Cobb-Douglas to Leontief	SI Text 4, Fig. S6
The ordering of fields by the value of targeting and of review survives changes in mean capability, the capability-resource association, the resource tail, pool size, the resource signal, and the review-signal model; the inequality of the gaps organizes the results	SI Text 5, Tables S6 to S12, Fig. S7
Planning the timing of spending adds little beside choosing recipients, and exists only under complementarity	SI Text 6, Fig. S8; SI Text 4, Table S5

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Model details. Output in each round is drawn from a Poisson distribution with rate $\lambda_i = AK_i R_i / (K_i + R_i)$, independently across researchers and rounds given the state; the units of output may be read as publications, findings, or units of scientific value. The production function is proportional to the harmonic mean of capability and resources. The harmonic mean makes the two inputs complements without making either an absolute requirement: a capable researcher with few

resources still produces, by working more slowly or borrowing equipment, but no surplus of one input fully substitutes for a shortage of the other. SI Text 4 repeats the central comparisons for a family of production functions from Cobb-Douglas to Leontief.

Between rounds, capability grows by $\epsilon \Lambda(K_i, R_i)$ with $\Lambda = K_i R_i / (K_i + R_i)$: doing research builds expertise, the same bottleneck that limits output limits growth, and capability never declines. Growth follows expected output, so a researcher's accumulation depends on their capability and resources and not on luck in a given round. Resources do not accumulate. The baseline R_{i0} recurs every round and represents the time and support a researcher has independent of the funder; a grant is consumed in the round it is given, and whatever lasts from it (skills, publications, standing) enters the model as capability.

The funder's total budget over the horizon is $b n \mathbb{E}[R_0]$, where $\mathbb{E}[R_0]$ is the expected baseline resources of one researcher, so at $b = 1$ the budget equals the community's expected baseline resources for one round. The budget does not grow with the horizon: a longer horizon varies the timing of spending, never its amount. Strategies that do not plan ahead spend $1/T$ of the budget each round; forward-looking strategies re-plan the entire remaining budget over the remaining horizon each round and execute the current round's allocation. For the timing runs (Fig. S8A and Fig. S4) the budget is scaled against capability instead, $b n \mathbb{E}[K_0]$, so that a community with few resources does not zero the budget. The simulations set the productivity constant $A = 1$; the code parameterizes the budget as $2 b' n \mathbb{E}[R_0]$, so its b' is half of the b reported here.

The funder knows the distributions of capability and resources but observes no researcher's capability or resources directly. It observes each researcher's output as it accumulates. A resource signal, observed once at the start, gives a noisy estimate of each researcher's baseline resources (noise τ_R), as one might obtain from institutional information. A review signal, also observed once at the start by the strategies that use it, gives a noisy estimate of capability (noise τ_K), as one might obtain from proposal evaluation and expert judgment; the smaller τ_K , the more informative the review. Formally, the resource signal is $R_{i0} + \eta_i$ with $\eta_i \sim N(0, \tau_R^2)$, and the review signal is $K_i + \xi_i$ with $\xi_i \sim N(0, \tau_K^2)$, independent across researchers; both are held fixed over the horizon, so a reviewer's misjudgment of a researcher persists across rounds, and only the output record accumulates. SI Text 5 reports the variant in which review is redrawn every round, and the variant in which review noise is multiplicative. Realized output is Poisson with mean λ_i . Bayesian strategies hold a posterior over each researcher's (K_i, R_{i0}) , computed from these likelihoods by importance sampling with $M = 200$ particles

Strategy	Information	Planning
No funding	none	none
Uniform	none	none
Track record	track record	none
Myopic, without review	track record + resource signal	current round
Forward, without review	track record + resource signal	remaining horizon
Myopic, with review	track record + resource signal + review	current round
Forward, with review	track record + resource signal + review	remaining horizon

TABLE S1. The funding strategies compared.

drawn afresh from the prior each round and weighted by the full likelihood of the record and signals to date (the convergence of the results in M is checked below), update it after each round, and allocate each dollar to the researcher whose expected output, given the posterior, rises most with it, by exact water-filling on the posterior expected marginal values. The forward-looking funder is a certainty-equivalent receding-horizon planner: each round it chooses the current allocation and a plan for the remaining budget to maximize expected output over the remaining horizon, treating future output as equal to its expectation, and re-plans after observing the round’s output; it does not value the information its own grants will generate (SI Text 6). The complete-information benchmark applies the gap rule of Proposition 1 to each round’s equal share of the budget with the true capabilities and resources; SI Text 1 compares it with the exact dynamic optimum.

Strategies. Table S1 lists the strategies compared. The Bayesian strategies differ in information, with or without the review signal, and in planning, myopic (optimizing the current round’s expected output) or forward-looking (optimizing expected output over the remaining horizon). Three baselines complete the set: no funding, uniform funding, which splits each round’s budget equally, and track-record funding, which allocates in proportion to the previous round’s output. Seed grants and lotteries modify these strategies as described in the main text.

Parameter ranges and defaults. The tail parameters α_K and α_R range from 1.3 to 3.5. Empirical distributions of research productivity have Pareto tail parameters near 1 to 2 (Lotka 1926; Nicholls 1989); since productivity is an imperfect guide to capability (main text), the range extends well beyond those estimates toward more equal fields. The budget scale b ranges from 0.01 to 3, from a funder whose budget is small beside the community’s own resources to one able to relieve most researchers’ bottlenecks, with larger budgets in the timing analyses (SI Text 6). The compounding rate ϵ ranges from near zero, where funding raises only current output, to 0.85, where this round’s funding substantially raises later capability. The review noise τ_K ranges from 0.05, nearly perfect review, to 20,

nearly uninformative review, a range wide enough to contain the noise implied by empirical estimates of review’s predictive accuracy. The copula parameter ρ ranges from -0.5 to 0.8 (rank correlations -0.48 to 0.79), and the horizon T from one round to ten. Capability and resources are coupled through a Gaussian copula: standard normal draws with correlation ρ are mapped through the normal distribution function to uniforms and then through the Pareto quantile functions. A Pearson correlation between two Pareto variables is undefined when either tail parameter is at most 2, so ρ is a copula parameter and not a correlation of the draws. Below a tail parameter of 2 the Pareto distribution has infinite variance, and sample means converge slowly: at $\alpha_K = 1.3$ with the scale set for a mean of 2, the mean capability across 100 populations of 50 is 1.77 and the median population has mean 1.48, because the rare draws that carry the mean are absent from most samples. For that reason the heavy-tailed results are reported with standard errors across populations, and Table S2 reports medians beside means. The rank correlation induced by the copula is $(6/\pi) \arcsin(\rho/2)$. Unless a figure’s settings below say otherwise: $T = 2$; $n = 50$; capability and baseline resources drawn from Pareto distributions with scale 1 and $\alpha_K = \alpha_R = 2$; $\rho = 0$; $\epsilon = 0.1$; $b = 1$; $\tau_K = \tau_R = 1$; seed-grant fraction 0.25 where seed grants are used.

Particles. Bayesian funders compute each researcher’s posterior by importance sampling: M particles (K, R_0) are drawn from the prior afresh in each round and weighted by the full likelihood of the researcher’s record and signals to date. With few particles the posterior can rest on a handful of them, and rare regions of the prior, where the most capable researchers of a heavy-tailed field sit, can be missed. Table S2 checks this. Raising M from 200 to 1,000 raises the value of review by 0.01 to 0.02 in every field at every noise level (the posteriors at $M = 200$ understate the capability of the most capable researchers), and raising it further to 5,000 changes the value of review by at most 0.005, within one standard error everywhere. The value of targeting uses no particles. Fig. 2B and C, the sensitivity analysis of SI Text 5, and Fig. 4 use $M = 1,000$; the remaining simulation figures use $M = 200$, and by the table their values of review are understated by about 0.01 to 0.02 of the records-only shortfall, which does not change any comparison they support.

Scoring and comparison. Each strategy is scored by the sum of expected outputs along its run, which equals realized output averaged over repetitions of the same allocations; this removes simulation noise from comparisons without favoring any strategy. Every configuration is run on independently drawn populations, with the same random draws (populations, signals, and realized output) used for every strategy, so that comparisons between strategies are paired. Results are

TABLE S2. Convergence in the number of particles. The value of review (share of the records-only shortfall recovered; mean over 100 populations, with the median in parentheses) and the effective sample size of the round-one posteriors (median across researchers and populations, with the tenth percentile in parentheses), by field, review noise, and number of particles M . Two rounds, $b = 1$, mean capability 2; the same 100 populations at every M . The value of targeting uses no particles and is unchanged.

α_K	τ_K	$M = 200$	$M = 1000$	$M = 5000$
<i>Value of review, mean (median)</i>				
1.3	0.3	0.91 (0.94)	0.94 (0.96)	0.94 (0.97)
1.3	1	0.77 (0.83)	0.79 (0.84)	0.79 (0.85)
1.3	3	0.48 (0.56)	0.50 (0.58)	0.50 (0.53)
2	0.3	0.84 (0.87)	0.86 (0.89)	0.87 (0.89)
2	1	0.63 (0.67)	0.65 (0.69)	0.65 (0.69)
2	3	0.28 (0.26)	0.30 (0.26)	0.29 (0.27)
3.5	0.3	0.62 (0.65)	0.63 (0.66)	0.64 (0.67)
3.5	1	0.31 (0.32)	0.32 (0.33)	0.32 (0.33)
3.5	3	0.05 (0.04)	0.06 (0.05)	0.06 (0.04)
<i>Effective sample size, median (tenth percentile)</i>				
1.3	0.3	81 (5)	396 (28)	1998 (136)
1.3	1	122 (15)	618 (73)	3081 (369)
1.3	3	142 (36)	711 (164)	3559 (796)
2	0.3	74 (6)	368 (33)	1843 (166)
2	1	124 (18)	616 (90)	3079 (447)
2	3	146 (46)	732 (227)	3667 (1137)
3.5	0.3	92 (12)	460 (56)	2300 (294)
3.5	1	136 (39)	685 (193)	3417 (955)
3.5	3	153 (67)	771 (314)	3857 (1558)

reported as shares of a named comparison, stated in each figure: most often the research output that funding adds (a strategy’s expected output minus the expected output under no funding), and otherwise the records-only shortfall or the optimal allocation’s gain. Standard errors, where shown, are across simulated populations.

Settings by figure. Table S3 lists, for each figure, the settings that differ from the defaults.

Code. The model, the sweeps, and the scripts that produce every figure and number in the paper are in the project repository; the archived version will be cited at publication.

SI Text 1. DERIVATION OF THE GAP RULE

The within-round allocation problem. A population of n researchers. Researcher i has capability $K_i > 0$ and baseline resources $R_{i0} \geq 0$, both fixed within

Figure	Settings that differ from the defaults
Fig. 2C	Review noise τ_K from 0.05 to 20; $\alpha_K \in \{1.3, 2, 3.5\}$ with the Pareto scale set to $2(\alpha_K - 1)/\alpha_K$ so that mean capability is 2 in every field; 100 populations per point; $M = 1,000$. The records-only shortfall relative to complete information is 44, 26, and 11 percent of the research output that complete-information funding adds, at the three tail parameters.
Fig. 2B	$T = 20$; 50 populations. Records-only is the myopic funder without review; with review is the myopic funder with a single review score observed at the start; $M = 1,000$. The shortfall is the funder's expected output below the complete-information optimum in that round, as a share of the optimum's gain over no funding. Under heavy-tailed capability ($\alpha_K = 1.3$, mean 2) the records-only shortfall falls from 60 to 14 percent against 6 to 2 percent with review, and the grant correlations run 0.12 to 0.55 against 0.89 in round one.
Fig. S3	True noise $\tau_K = 3$; believed noise from 0.1 to 10; $\alpha_K \in \{1.3, 2\}$; 200 populations. Each negative value is within two standard errors of zero.
Fig. S4	$T = 6$; community with few resources (Pareto scale for resources 0.001); $\alpha_K = 1.3$; no review signal ($\tau_K = 100$); $\epsilon = 10^{-4}$; budget scaled against capability; $b = 1$ (50 populations) and $b = 6$ (24 populations, standard error of each share at most 0.006).
Fig. S8A	$T = 5$; forward-looking funder with review; budget scaled against capability; resource scales 0.001, 0.3, and 3; ϵ from 10^{-4} to 0.85; 200 populations. Five resource scales were run; the figure shows three, and the other two lie between them.
Fig. S8B	T from 2 to 10 (to 5 for $\epsilon \in \{0.05, 0.15, 0.55\}$); $\epsilon \in \{0.05, 0.15, 0.3, 0.55, 0.85\}$; 64 to 200 populations per point. The gain is the forward-looking funder's output over the myopic funder's, both with review; negative gains, all within noise of zero, are shown as zero.
Fig. 3A	Review-informed forward-looking funder; seed-grant fraction from 0 to 0.75; heavy-tailed field ($\alpha_K = 1.3$) with reliable review ($\tau_K = 0.3$) at $b \in \{0.2, 1, 2\}$, and the default field at $b = 1$; 200 populations. The output lost is reported as a share of the review-informed funder's gain over no funding in the same setting.
Fig. 3B	Single round; $n = 50$; a fifth of the population funded; review score K_i plus Gaussian noise of standard deviation τ ; expected output computed exactly given each allocation; 2,000 populations per point (Monte Carlo standard error at most 0.002). Left: $\alpha_K = 1.3$, $b = 0.2$. Right: $\alpha_K = 3.5$, $b = 1$.
Fig. S6, Table S5	Production functions with $\gamma \in \{0, -3\}$ and Leontief; 100 populations per point (50 for Leontief); $T = 2$ for the signal value and $T = 5$ for the timing table. These runs use the greedy allocator, the only one implemented for the general family (the harmonic-mean runs use exact water-filling), so their magnitudes are compared with one another and not with the main-text figures. Leontief allocations use a finer allocation step, with ties at the kink broken by fill order.
Fig. S5	Single round; review-informed funder; γ from 0 to -12 and Leontief; three field-and-budget combinations ($\alpha_K = 1.3$, $b = 0.2$; $\alpha_K = 2$, $b = 2$; $\alpha_K = 3.5$, $b = 2$); 200 populations per point, 400 at the refinement points of the evenly spread curve.
Fig. S7, Tables S6 to S12	$T = 2$; $b = 1$; myopic funders with and without review and the complete-information benchmark; $\rho \in \{-0.5, 0, 0.5, 0.8\}$; $\alpha_R \in \{1.3, 2, 3.5\}$; $n \in \{25, 50, 100, 200\}$; Pareto scale 1 or set to give mean capability 2; review score $K_i + \beta R_{i0}$ plus noise with $\beta \in \{0, 0.25, 0.5, 1\}$; resource signal noise $\sigma \in \{0.1, 0.2, 1, 2, 10\}$; and

the round. A grant $g_i \geq 0$ adds to researcher i 's resources for the round, so that their expected output is $F(K_i, R_{i0} + g_i)$. We prove the gap rule for every production function of the form

$$F(K, R) = A K h(R/K),$$

where $A > 0$ is the productivity constant and $h: [0, \infty) \rightarrow [0, \infty)$ is continuous, continuously differentiable on $(0, \infty)$, strictly increasing, and strictly concave, so that h' is positive and strictly decreasing. This is the class of production functions with constant returns to scale in (K, R) and diminishing returns to resources. The harmonic mean of the main text is the instance $h(x) = x/(1+x)$, for which $F(K, R) = AKR/(K+R)$; every member of the power-mean family of SI Text 4 with exponent $\gamma < 1$ is an instance, with $h(x) = ((1+x^\gamma)/2)^{1/\gamma}$. Write

$$\varphi_i(g) = A K_i h\left(\frac{R_{i0} + g}{K_i}\right)$$

for researcher i 's expected output as a function of their grant alone; for the harmonic mean, $\varphi_i(g) = AK_i(R_{i0} + g)/(K_i + R_{i0} + g)$.

Throughout, the round's budget is $B > 0$. Nothing else about the model is used, so the result holds for every realized population and every round's budget; across rounds, the constant c and the funded set are recomputed from the new population state.

The funder observes $(K_i, R_{i0})_{i=1}^n$ and divides the budget so as to maximize expected output:

$$(P) \quad \max_{g_1, \dots, g_n \geq 0} \sum_{i=1}^n \varphi_i(g_i) \quad \text{subject to} \quad \sum_{i=1}^n g_i \leq B.$$

Three lemmas.

Lemma 1 (the return to a grant). *For each i , on $g \in [0, \infty)$: (i) φ_i is strictly increasing, with $\varphi'_i(g) = A h'((R_{i0} + g)/K_i) > 0$; (ii) φ_i is strictly concave, since φ'_i is strictly decreasing in g ; (iii) $\varphi_i(g) \uparrow AK_i \bar{h}$ as $g \rightarrow \infty$, where $\bar{h} = \lim_{x \rightarrow \infty} h(x) \leq \infty$. For the harmonic mean, $\varphi'_i(g) = AK_i^2/(K_i + R_{i0} + g)^2$, $\varphi''_i(g) = -2AK_i^2/(K_i + R_{i0} + g)^3$, $\varphi'_i(g) < A$ whenever $R_{i0} + g > 0$, and $\bar{h} = 1$, so expected output never exceeds AK_i .*

Proof. Differentiating $\varphi_i(g) = AK_i h((R_{i0} + g)/K_i)$ in g gives (i), and (ii) follows because h' is strictly decreasing and its argument increases with g . For (iii), h is increasing, so $h(x) \uparrow \bar{h}$ as $x \rightarrow \infty$. For the harmonic mean, write $S = K_i + R_{i0} + g$; then $\varphi_i(g) = AK_i - AK_i^2/S$, and the stated derivatives follow (note $dS/dg = 1$); $\varphi'_i(g) = AK_i^2/S^2 < A$ exactly when $S > K_i$, that is, when $R_{i0} + g > 0$. $\square$

Lemma 1 says that the marginal value of a dollar to researcher i depends on capability and resources only through their ratio, decreases as that ratio rises, and so increases with capability and decreases with the resources they already hold; that each dollar granted lowers the value of the next; and that, when h is bounded, expected output never exceeds a ceiling set by capability.

Lemma 2 (existence, uniqueness, exhaustion). *Problem (P) has exactly one solution $g^* = (g_1^*, \dots, g_n^*)$, and it exhausts the budget: $\sum_i g_i^* = B$.*

Proof. The feasible set is compact and convex and the objective $f(g) = \sum_i \varphi_i(g_i)$ is continuous and, by lemma 1(ii), strictly concave, so a maximizer exists and is unique. If $\sum_i g_i^* < B$, increasing any one g_i^* slightly stays feasible and raises f (lemma 1(i)), contradicting optimality. $\square$

Lemma 3 (equal marginal value). *A feasible allocation g with $\sum_i g_i = B$ solves (P) if and only if there is a $\nu > 0$ such that*

$$\varphi'_i(g_i) = \nu \text{ for every } i \text{ with } g_i > 0, \quad \varphi'_i(0) \leq \nu \text{ for every } i \text{ with } g_i = 0,$$

where $\varphi'_i(0)$ may be $+\infty$ (as it is when $R_{i0} = 0$ and $h'(0^+) = \infty$), in which case researcher i is funded.

Proof. Necessity. Let g^* solve (P). Since $B > 0$ is exhausted (lemma 2), some researcher is funded: $g_i^* > 0$ for some i . Fix any such i and any $j \neq i$, and for $0 < \delta \leq g_i^*$ consider the transfer that replaces g_i^* by $g_i^* - \delta$ and g_j^* by $g_j^* + \delta$, leaving the rest unchanged. The result is feasible, so optimality gives

$$\varphi_j(g_j^* + \delta) - \varphi_j(g_j^*) \leq \varphi_i(g_i^*) - \varphi_i(g_i^* - \delta).$$

Dividing by δ and letting $\delta \downarrow 0$ yields $\varphi'_j(g_j^*) \leq \varphi'_i(g_i^*)$. If j is also funded, the same argument with i and j exchanged gives the reverse inequality, so all funded researchers share a common marginal value; call it $\nu = \varphi'_i(g_i^*) > 0$ (positivity by lemma 1(i)). For unfunded j , the displayed inequality reads $\varphi'_j(0) \leq \nu$.

Sufficiency. Let g satisfy the condition with multiplier ν , and let $\tilde{g}$ be any feasible allocation. Concavity of φ_i gives, for every i , $\varphi_i(\tilde{g}_i) \leq \varphi_i(g_i) + \varphi'_i(g_i)(\tilde{g}_i - g_i)$. For funded i , $\varphi'_i(g_i) = \nu$, so $\varphi'_i(g_i)(\tilde{g}_i - g_i) = \nu(\tilde{g}_i - g_i)$. For unfunded i , $g_i = 0$ and $\tilde{g}_i - g_i = \tilde{g}_i \geq 0$, so $\varphi'_i(0)(\tilde{g}_i - g_i) \leq \nu(\tilde{g}_i - g_i)$. Summing over i :

$$f(\tilde{g}) \leq f(g) + \nu \sum_i (\tilde{g}_i - g_i) = f(g) + \nu \left(\sum_i \tilde{g}_i - B \right) \leq f(g),$$

since $\tilde{g}$ is feasible and $\nu > 0$. So g solves (P). $\square$

Lemma 3 is the formal content of the sentence in the main text: an allocation is optimal exactly when no dollar can be moved to a researcher who can produce more value with it.

The proposition.

Proposition 1 (the gap rule). *Let $F(K, R) = A K h(R/K)$ with h increasing and strictly concave. Then problem (P) has the unique solution*

$$g_i^* = \max(c K_i - R_{i0}, 0), \quad i = 1, \dots, n,$$

where $c > 0$ is the unique constant satisfying the budget identity $\sum_{i=1}^n \max(c K_i - R_{i0}, 0) = B$. Equivalently, $c = (h')^{-1}(\nu/A)$, where ν is the common marginal value of lemma 3; for the harmonic mean, $c = \sqrt{A/\nu} - 1$ with $\nu \in (0, A)$.

Proof. Let g^* be the unique solution (lemma 2) and ν the multiplier that lemma 3 associates with it. The proof proceeds in four steps.

Step 1: $c = (h')^{-1}(\nu/A)$ is well defined and positive. Some researcher i is funded, and for them lemma 3 gives $\nu = \varphi'_i(g_i^*) = A h'((R_{i0} + g_i^*)/K_i)$, so ν/A lies in the range of h' ; since h' is strictly decreasing, its inverse is defined there, and $c = (h')^{-1}(\nu/A) = (R_{i0} + g_i^*)/K_i > 0$ because $g_i^* > 0$. For the harmonic mean, $h'(x) = 1/(1+x)^2$, so $(h')^{-1}(y) = 1/\sqrt{y} - 1$ and $c = \sqrt{A/\nu} - 1$; positivity of c is then $\nu < A$, which lemma 1 gives directly.

Step 2: funded researchers receive exactly their gap. Let $g_i^* > 0$. By lemma 3, $A h'((R_{i0} + g_i^*)/K_i) = \nu$, so $(R_{i0} + g_i^*)/K_i = (h')^{-1}(\nu/A) = c$ and $g_i^* = c K_i - R_{i0}$. Since $g_i^* > 0$, this equals $\max(c K_i - R_{i0}, 0)$. Every funded researcher is thus brought to the same ratio of resources to capability, c .

Step 3: unfunded researchers have no gap. Let $g_i^* = 0$. By lemma 3, $A h'(R_{i0}/K_i) = \varphi'_i(0) \leq \nu = A h'(c)$, and since h' is strictly decreasing, $R_{i0}/K_i \geq c$, that is, $R_{i0} \geq c K_i$. Then $\max(c K_i - R_{i0}, 0) = 0 = g_i^*$.

Steps 2 and 3 establish $g_i^* = \max(c K_i - R_{i0}, 0)$ for every i , and the budget identity is exhaustion (lemma 2). It remains to show c is the only constant satisfying that identity.

Step 4: the budget identity pins c uniquely. Define $G(c) = \sum_{i=1}^n \max(c K_i - R_{i0}, 0)$ for $c \geq 0$. G is continuous and nondecreasing, with $G(0) = 0$ and $G(c) \rightarrow \infty$ as $c \rightarrow \infty$. Moreover G is strictly increasing wherever it is positive: if $G(c) > 0$, then $c K_j > R_{j0}$ for some j , and for any $c' > c$, $G(c') \geq G(c) + K_j (c' - c) > G(c)$. Now suppose $c_1 < c_2$ both satisfy $G(c) = B$. Since $B > 0$, $G(c_1) > 0$, so $G(c_2) > G(c_1) = B$, a contradiction. $\square$

The proof uses only that the marginal value of a dollar depends on K and R through the ratio R/K and falls as that ratio rises. Constant returns to scale give the first property and diminishing returns to resources the second; the harmonic form contributes nothing beyond the closed form of c . The constant c is the ratio of resources to capability at which the optimal allocation leaves every funded researcher, and so also the rate at which the rule converts a unit of capability into a target level of resources.

Consequences.

Corollary 1 (funding frontier and budget comparative statics). *(i) Researcher i is funded, $g_i^* > 0$, exactly when $K_i > R_{i0}/c$. (ii) As a function of the budget B , the constant $c(B)$ is continuous and strictly increasing; hence the funded set is nondecreasing in B , every funded researcher's grant $c(B)K_i - R_{i0}$ is strictly increasing in B , and the common marginal value $\nu(B) = Ah'(c(B))$ is strictly decreasing in B . Both parts follow from the formula and Step 4.*

Corollary 2 (the value of targeting vanishes in the budget). *Let $f^*(B)$ denote the optimal value of (P) and $f^u(B) = \sum_i \varphi_i(B/n)$ the value of uniform funding. Then $0 \leq f^*(B) - f^u(B)$ for every B . If h is bounded, $\bar{h} < \infty$, then $f^*(B) - f^u(B) \rightarrow 0$ as $B \rightarrow \infty$.*

Proof. The first inequality holds because uniform funding is feasible. For the second: by lemma 1(iii), $\varphi_i(g) < AK_i\bar{h}$ for every g , so $f^*(B) \leq A\bar{h}\sum_i K_i$; and $f^u(B) = \sum_i \varphi_i(B/n) \rightarrow A\bar{h}\sum_i K_i$ as $B \rightarrow \infty$. Hence $0 \leq f^*(B) - f^u(B) \leq A\bar{h}\sum_i K_i - f^u(B) \rightarrow 0$. $\square$

Two remarks. First, the corollary says nothing about monotonicity: $f^*(B) - f^u(B)$ is zero at $B = 0$ and positive for small $B > 0$ whenever researchers differ, so it cannot decrease throughout. The decrease reported in the main text is a numerical finding about the ratio $(f^* - f^u)/(f^u - f^0)$, with f^0 the output under no funding: it fell monotonically in the budget, on budget scales from 0.02 to 3, in every one of 300 populations of 50 researchers drawn from the default distributions, and in the population of Fig. S2. Since uniform funding's own gain over no funding does not vanish (it approaches $A\bar{h}\sum_i K_i - \sum_i \varphi_i(0) > 0$), the ratio vanishes with the difference. Second, boundedness of h is what makes targeting's value vanish: in the power-mean family, h is bounded exactly when $\gamma < 0$, with $\bar{h} = 2^{-1/\gamma}$ (so $\bar{h} = 1$ for the harmonic mean), whereas under Cobb-Douglas production ($\gamma \rightarrow 0$, $h(x) = \sqrt{x}$) output has no ceiling set by capability and $f^*(B) - f^u(B)$ grows without bound in B for any population with unequal capabilities. The Leontief form $h(x) = \min(1, x)$ is the limit $\gamma \rightarrow -\infty$ of the family; it is not strictly concave,

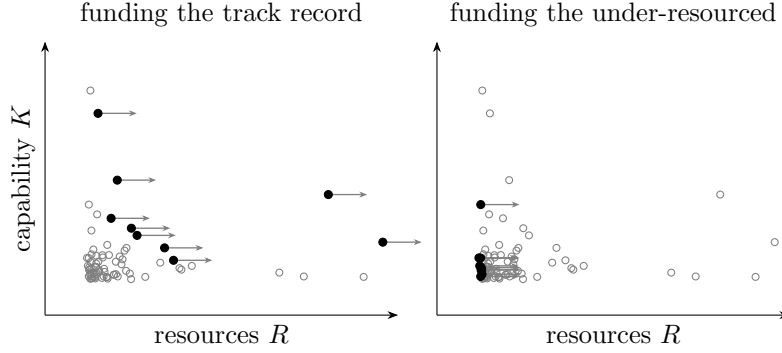


FIGURE S1. The two intuitive schemes on the population of Fig. 1, with the budget of that figure's smaller frontier divided equally among nine researchers. Left: funding the track record gives the grants (arrows) to the nine researchers with the highest expected output; most of them are well resourced and of middling capability, so an added dollar adds little. Right: funding the under-resourced gives the grants to the nine researchers with the fewest resources, chosen without regard to capability, so most of the money goes to researchers of low capability. On this population the gap rule adds 4.0 units of expected output over no funding; funding the track record adds 2.0, funding the under-resourced 2.1, and dividing the budget uniformly across all 75 researchers 2.0.

so the proposition does not apply to it directly, and it is treated numerically in SI Text 4.

Example 1 (funding the track record can produce less than uniform funding). *Two researchers, harmonic production, $A = 1$, budget $B = 2$. Researcher 1: $K_1 = 8$, $R_{10} = 8$ (productive and well resourced; expected output $\varphi_1(0) = 4$). Researcher 2: $K_2 = 4$, $R_{20} = 1$ (capable but with few resources; $\varphi_2(0) = 4/5$). Allocating in proportion to expected output gives $g = (5/3, 1/3)$ and total output $285/53 \approx 5.38$. Uniform funding, $g = (1, 1)$, gives $284/51 \approx 5.57$. The gap rule ($c = 3/4$) gives $g^* = (0, 2)$ and $40/7 \approx 5.71$.*

Example 2 (funding the under-resourced can produce less than uniform funding). *Two researchers, harmonic production, $A = 1$, budget $B = 2$. Researcher 1: $K_1 = 10$, $R_{10} = 5$. Researcher 2: $K_2 = 1/2$, $R_{20} = 0$ (the scarcest resources in the population). Directing the budget to the researcher with the fewest resources gives total output $56/15 \approx 3.73$. Uniform funding gives $49/12 \approx 4.08$. The gap rule ($c = 2/3$) gives $g^* = (5/3, 1/3)$ and $21/5 = 4.2$.*

The one-round rule and the sequential optimum. Proposition 1 maximizes one round's expected output. Over T rounds a grant also raises the recipient's later capability, by $\epsilon \Lambda(K_i, R_{i0} + g_i)$ per round, so the allocation that maximizes

expected output over the horizon can differ from the gap rule in two ways: in whom it funds within a round, since a dollar’s future value depends on the recipient’s growth, and in how it divides the budget across rounds. Because capability growth follows expected output, the complete-information problem over T rounds is a deterministic program, and for $T = 2$ we solve it exactly: the round-1 grants and the round-1 share of the budget are chosen freely (a softmax parameterization over the n recipients and a logistic share, optimized by quasi-Newton iteration from the gap rule as the starting point), and the remaining budget is allocated in round 2 by the gap rule on the grown capabilities, which is optimal for the last round. Table S4 compares this optimum with the benchmark used throughout the paper, the gap rule applied to half the budget in each round, on 20 populations of 50 at each setting, with mean capability 2. For $T = 5$ we restrict the comparison to the division of the budget across rounds, with the gap rule choosing recipients in each round, and compare the best division with the even one.

The benchmark loses at most 1.4 percent of the research output that funding adds in any population, and under 1 percent on average in every setting; its grants correlate with the optimum’s at 0.98 or above. The loss grows with the compounding rate and with the budget, is larger where capability is evenly spread, and is indistinguishable from zero at the default compounding rate. The dynamic optimum spends 36 to 49 percent of the budget in the first round, because the capability that round-1 output builds makes round-2 dollars more productive; the same shift appears in the forward-looking funder of SI Text 6. Over five rounds, with the gap rule choosing recipients, the even division of the budget across rounds loses at most 4.0 percent of what the best division adds at the highest compounding rate, and at most 0.1 percent at the default rate, and the best division shifts spending later (center of mass 0.51 to 0.65). The one-round rule is therefore the right benchmark for the sequential problem at the compounding rates we study, and the value of information reported in the main text is measured against a benchmark within about one percent of the sequential optimum.

SI Text 2. LOTTERIES AND EQUAL DIVISION

Fix a round, a budget $B > 0$, and a pool $P \subseteq \{1, \dots, n\}$ of m researchers. The *equal division* over P grants each researcher in P the amount B/m . A *lottery* over P with $k \leq m$ winners selects k researchers from P uniformly at random and grants each winner B/k . Expected output is evaluated as in SI Text 1: given an allocation g , it is $\sum_i \varphi_i(g_i)$, and for a random allocation, the expectation is taken over the draw as well.

Proposition 2 (lotteries are outperformed by equal division). *For every pool P , budget B , and winner count $k \leq m$, the lottery’s expected output does not exceed*

TABLE S4. The round-by-round gap rule against the sequential optimum under complete information. Two rounds: the output lost by applying the gap rule to half the budget in each round, as a percent of the research output that the exact two-round optimum adds over no funding (mean and maximum over 20 populations); the correlation between the two allocations' first-round grants (mean and minimum); and the share of the budget the optimum spends in round one. Five rounds: the output lost by an even division of the budget across rounds, as a percent of what the best division adds, with the gap rule choosing recipients in every round, and the center of mass of the best division (0.5 is even; higher is later). Mean capability 2; $n = 50$.

α_K	ϵ	b	two rounds, exact optimum			five rounds, division only		
			loss, % (max)	correlation (min)	round-1 share	even division loses, %	center of mass	
1.3	0.1	0.2	0.0 (0.0)	1.000 (1.000)	0.49	0.0	0.51	
1.3	0.1	1	0.0 (0.0)	1.000 (1.000)	0.49	0.1	0.51	
1.3	0.1	2	0.0 (0.0)	1.000 (1.000)	0.49	0.1	0.51	
1.3	0.5	0.2	0.1 (0.4)	0.999 (0.997)	0.46	0.4	0.54	
1.3	0.5	1	0.2 (0.6)	1.000 (0.999)	0.45	1.0	0.55	
1.3	0.5	2	0.3 (0.5)	1.000 (0.999)	0.45	1.4	0.55	
1.3	0.85	0.2	0.3 (1.1)	0.999 (0.994)	0.44	0.9	0.56	
1.3	0.85	1	0.6 (1.4)	0.999 (0.998)	0.42	2.1	0.57	
1.3	0.85	2	0.7 (1.2)	1.000 (0.997)	0.42	2.9	0.58	
2	0.1	0.2	0.0 (0.0)	1.000 (1.000)	0.49	0.0	0.51	
2	0.1	1	0.0 (0.0)	1.000 (1.000)	0.49	0.1	0.52	
2	0.1	2	0.0 (0.0)	1.000 (1.000)	0.49	0.1	0.52	
2	0.5	0.2	0.2 (0.4)	0.999 (0.997)	0.45	0.6	0.55	
2	0.5	1	0.3 (0.5)	1.000 (0.999)	0.44	1.2	0.57	
2	0.5	2	0.3 (0.4)	1.000 (0.999)	0.44	1.6	0.57	
2	0.85	0.2	0.4 (1.1)	0.997 (0.989)	0.41	1.2	0.58	
2	0.85	1	0.8 (1.2)	0.999 (0.996)	0.41	2.6	0.60	
2	0.85	2	0.8 (1.1)	0.999 (0.997)	0.41	3.3	0.60	
3.5	0.1	0.2	0.0 (0.0)	1.000 (0.999)	0.48	0.1	0.52	
3.5	0.1	1	0.0 (0.0)	1.000 (1.000)	0.49	0.1	0.52	
3.5	0.1	2	0.0 (0.0)	1.000 (1.000)	0.49	0.1	0.52	
3.5	0.5	0.2	0.3 (0.5)	0.997 (0.994)	0.41	1.3	0.60	
3.5	0.5	1	0.4 (0.4)	0.999 (0.998)	0.43	1.8	0.59	
3.5	0.5	2	0.4 (0.4)	0.999 (0.999)	0.44	1.9	0.58	
3.5	0.85	0.2	0.8 (1.2)	0.993 (0.985)	0.36	2.8	0.65	
3.5	0.85	1	0.9 (1.1)	0.997 (0.995)	0.39	3.7	0.63	
3.5	0.85	2	0.9 (1.0)	0.998 (0.997)	0.41	4.0	0.62	

the equal division's, with equality only if $k = m$. In particular, a lottery over the whole population, at any winner count, produces at most uniform funding's expected output.

Proof. Researchers outside P receive nothing under either scheme. For $i \in P$, the lottery's grant to i is a random variable $\tilde{g}_i$ equal to B/k with probability k/m

and 0 otherwise, so $\mathbb{E}[\tilde{g}_i] = B/m$. By linearity, the lottery's expected output is $\sum_{i \notin P} \varphi_i(0) + \sum_{i \in P} \mathbb{E}[\varphi_i(\tilde{g}_i)]$. Since φ_i is strictly concave (Lemma 1(ii)), Jensen's inequality gives $\mathbb{E}[\varphi_i(\tilde{g}_i)] \leq \varphi_i(\mathbb{E}[\tilde{g}_i]) = \varphi_i(B/m)$ for every $i \in P$, with equality only if $\tilde{g}_i$ is degenerate, that is, $k = m$. Summing over i yields the claim; taking $P = \{1, \dots, n\}$ yields the comparison with uniform funding. $\square$

Remark 1 (what the proposition does and does not cover). *The proposition compares expected output within a fixed pool. It measures the effect of the randomization alone: whatever pool review selects, replacing the lottery within that pool by the equal division of the same money weakly increases expected output. The proposition says nothing about the choice of pool, about the review ranking within the pool that the equal division also ignores, or about anything a lottery saves outside the model (review costs, proposal effort, bias).*

Remark 2 (minimum viable grant size). *The proposition needs divisible grants and diminishing returns at every grant size. Suppose instead that a project produces nothing unless its grant reaches a minimum m , and that $B < m|P|$ for a pool P . Equal division then gives every researcher in P less than m and produces no output, while a lottery with $k = \lfloor B/m \rfloor$ winners, each receiving $B/k \geq m$, produces the output of k funded projects. Where minimum project scale binds, the comparison of the proposition reverses, and this qualification applies wherever the main text compares lotteries with equal division.*

SI Text 3. EXTENDED RESULTS

The value of targeting decreases in the budget. The value of targeting is the expected output the optimal allocation adds over equal division of the same budget. By Corollary 2 of SI Text 1 it vanishes as the budget grows, for every production function under which output has a ceiling set by capability; that it also decreases throughout is a numerical finding, which held in every population we sampled (SI Text 1). Fig. S2 shows it for one population, as a fraction of uniform funding's gain over no funding. The same dependence on the budget sets how much output seed grants and lotteries give up and when the informed funder concentrates rather than spreads its grants (main text).

For a funder allocating with records and review rather than with complete information, the value of targeting relative to uniform funding's gain over no funding decreases from 1.17 to 0.72 across budget scales $b = 0.2$ to 2 with heavy-tailed capability ($\alpha_K = 1.3$) and reliable review ($\tau_K = 0.3$), and from 0.74 to 0.25 in the default field with the default signal ($\alpha_K = 2$, $\tau_K = 1$); two rounds, 200 simulated populations per point, the populations of Fig. 4A. Giving up targeting loses little

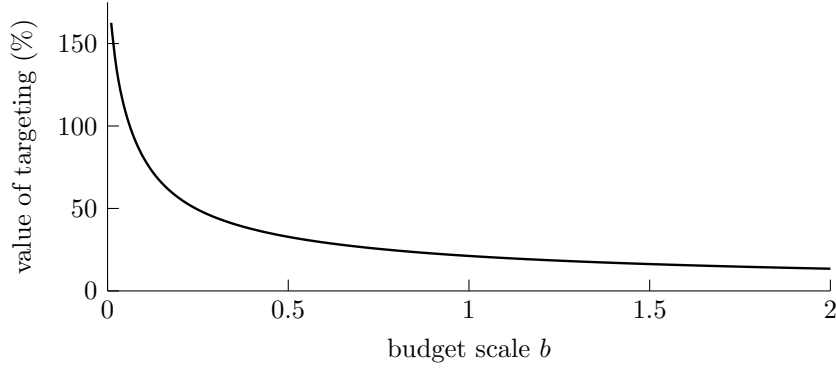


FIGURE S2. The value of targeting decreases in the budget. The additional expected output of the optimal allocation over uniform funding, as a fraction of uniform funding’s gain over no funding, against the budget scale b , for one population drawn from the default distributions. Corollary 2 proves that the value vanishes as the budget grows; the decrease throughout is a numerical finding (SI Text 1).

output under the same conditions that make review worth little: where capability is evenly spread, the budget is large, or review is uninformative.

Overtrusting review. The results on review’s value require the funder to weight review at its true informativeness. A funder that overtrusts review, treating a noisy signal as reliable, can do worse than one that ignores review altogether: with moderately spread capability, overtrust turns review’s value negative; with heavy-tailed capability it reduces the value without reversing it (Fig. S3). A funder that overtrusts tenfold (true $\tau_K = 3$, belief 0.3) loses more than review’s full value at the default capability distribution, whereas a funder that undertrusts tenfold (true $\tau_K = 0.3$, belief 3) retains over half of it. Overtrusting review loses more output than undertrusting it.

Grant size, learning, and the schedule. How much a funder can learn from its own grants depends on their size. A researcher on a small grant produces output nearly proportional to the grant and nearly independent of capability, so a funder whose grants are small learns almost nothing it can exploit later: in a field whose researchers have few resources, with no review signal, such a funder gains almost nothing over uniform funding. As grants approach the scale of capability itself, output separates the capable from the rest, and the funder recovers most of the value of discrimination, with timing that spends a small share early, so that output can be observed, and the rest once the funder has learned who is capable (Fig. S4). In a community whose researchers have few resources, with heavy-tailed capability and no review signal, over six rounds, the Bayesian funder’s gain over

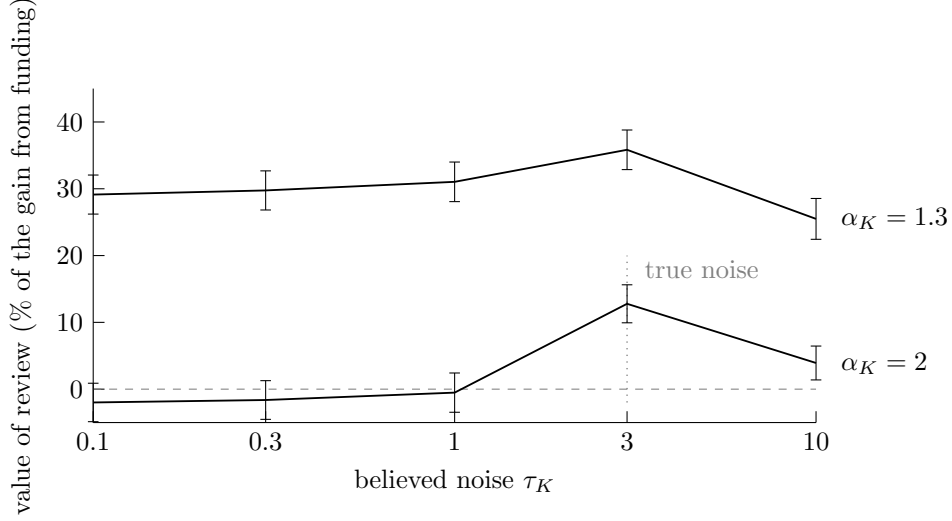


FIGURE S3. Overtrusting review loses more research output than undertrusting it. The value of review as a percent of the research output that a correctly calibrated review-informed funder adds over no funding, as the funder’s believed signal noise varies while the true noise is $\tau_K = 3$ (dotted rule); bars are one standard error. At the default capability distribution, overtrust takes the value below zero. Settings: SI Materials and Methods.

uniform funding increases from a third of one percent of uniform funding’s output at the default grant size to nine percent at six times that size and ten percent at twelve times, roughly the gain a reliable review signal delivers at the default size.

The shift of spending toward later rounds comes from the output researchers produce whether or not they are funded: that output compounds and reveals capability while the funder waits. Growth from grant-added output alone would shift spending slightly toward earlier rounds, since earlier grants have longer to compound. With growth from unfunded output switched off, the center of mass of spending is 0.48 to 0.49, below even; with growth from grant-added output switched off, it is 0.53 to 0.68 (five rounds).

Concentration and the production function. One might hope that the production function settles whether funding should be concentrated among a few researchers or spread across many: the harder it is to substitute resources for capability, the more the money should go to the few researchers who can use it. Across the family of production functions of SI Text 4, from Cobb-Douglas to Leontief, it does not (Fig. S5). With a tight budget in a heavy-tailed field, stronger complementarity does concentrate the review-informed funder’s grants: the Gini coefficient of its grants rises from 0.92 to 0.95. With an ample budget,

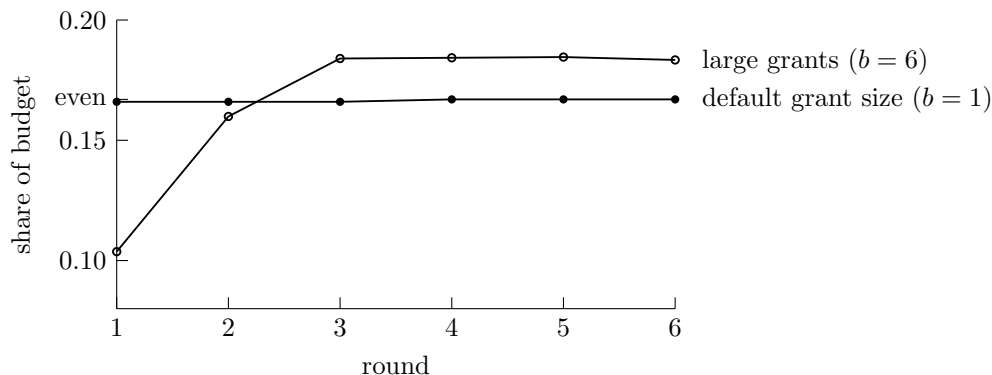


FIGURE S4. With large grants the funder spends a small share first and the rest once it has observed research output. The share of the budget spent in each of six rounds, in a field whose researchers have few resources, with heavy-tailed capability and no review signal, at the default grant size (filled) and at six times that size (open). Settings: SI Materials and Methods.

stronger complementarity spreads them, from 0.35 to 0.21 in the default field and from 0.63 to 0.45 in a heavy-tailed field, a pattern consistent with a funder that, once the budget allows, brings every researcher’s resources up toward the level their capability can use. Where capability is evenly spread and the budget ample, the relation is not even monotone: the Gini coefficient is 0.16 at Cobb-Douglas, peaks at 0.196 near $\gamma = -6$, and falls to 0.185 at $\gamma = -12$, 3.5 standard errors above the $\gamma = -12$ level. How widely funds should be spread is set jointly by the production function, the budget, and the field’s capability distribution; no one of the three decides it alone.

SI Text 4. ROBUSTNESS TO THE PRODUCTION FUNCTION

This appendix asks which of our results depend on the harmonic-mean production function of SI Materials and Methods. We embed that function in a family indexed by an exponent γ : expected output is $A M_\gamma(K, R)$, where $M_\gamma(K, R) = ((K^\gamma + R^\gamma)/2)^{1/\gamma}$ is the power mean of capability and resources. At $\gamma = -1$ the power mean is the harmonic mean $2KR/(K + R)$, which is our production function up to the constant A . In the limit $\gamma \rightarrow 0$ (Cobb-Douglas) it is the geometric mean $\sqrt{KR}$, and a shortfall in either input can be offset by more of the other; as γ decreases the inputs grow harder to substitute; in the limit $\gamma \rightarrow -\infty$ (Leontief) it is $\min(K, R)$, and output depends only on the scarcer input. We re-run the model’s central comparisons across this family. Settings: SI Materials and Methods.

The value of the review signal survives the whole family (Fig. S6). At every production function tested, from Cobb-Douglas to Leontief, the signal’s value increases

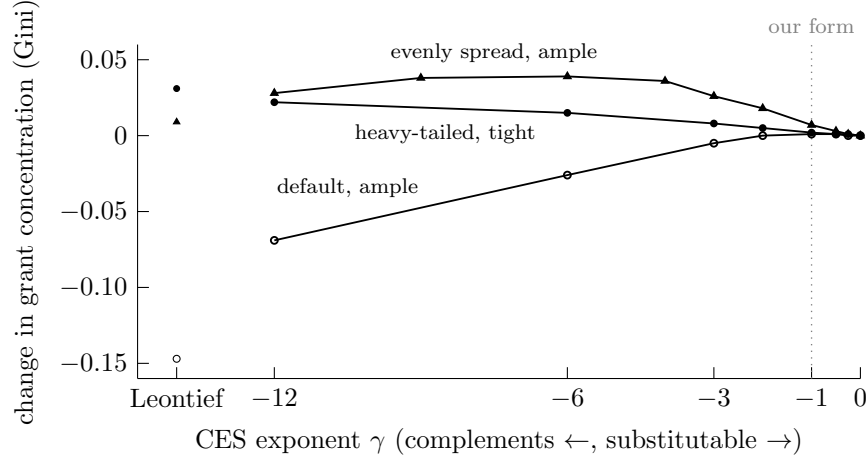


FIGURE S5. Complementarity concentrates funding only when the budget is small. The change in the Gini coefficient of the review-informed funder’s grants, relative to its Cobb-Douglas level, as the production function moves from Cobb-Douglas ($\gamma = 0$) toward Leontief (detached marks at the left edge), for three field-and-budget combinations. Settings: SI Materials and Methods.

as capability grows more heavy-tailed and as review grows more informative. The ordering of review’s value by field (main text) is therefore not an artifact of our choice of production function. What changes with the production function is the scale. The more complementary the inputs, the more review is worth: under heavy-tailed capability with reliable review, the signal’s value increases from about 16 percent of the research output that funding with review adds at Cobb-Douglas to about 47 percent at $\gamma = -3$ and about half at Leontief (15.8, 47.1, and 51.4 percent). Cobb-Douglas lies below the two more complementary production functions at every tested tail and noise level at which the value is distinguishable from zero; $\gamma = -3$ and Leontief are within one standard error of each other at the noisiest review under heavy tails. Complementarity raises what targeting is worth, and with it the value of the information that targeting requires.

The timing results of SI Text 6 do not survive the substitutable end of the family (Table S5). Under Cobb-Douglas production the gain from planning ahead is indistinguishable from zero at every compounding rate, and the forward-looking funder’s timing stays within about a hundredth of even. At $\gamma = -3$, a production function more complementary than ours, both the gain from planning ahead and the shift of spending toward later rounds increase with the compounding rate and exceed their Cobb-Douglas values at every rate. The timing results hold where capability and resources are complements and vanish where they are substitutable.

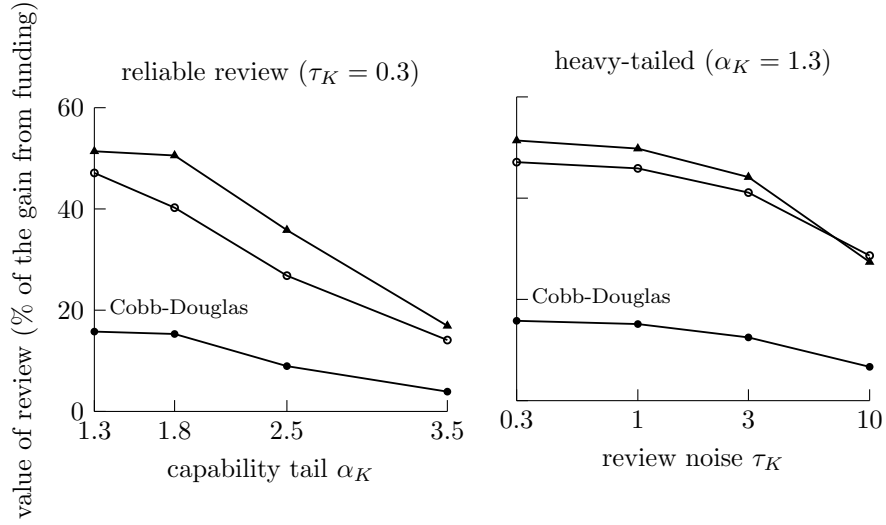


FIGURE S6. The structure of review’s value survives the production function. The value of the review signal, as a percent of the research output that funding with review adds over no funding, across the family of production functions: at every production function tested, the value increases as capability grows more heavy-tailed (left) and as review grows more informative (right), and its scale increases with complementarity. Filled circles: Cobb-Douglas; open circles: $\gamma = -3$; triangles: Leontief. Settings: SI Materials and Methods.

How widely the informed funder spreads its grants across this family is reported in the main text (Fig. S5).

SI Text 5. SENSITIVITY OF THE VALUE OF TARGETING AND OF REVIEW

The main text varies the tail of the capability distribution, the budget, and review noise, and holds fixed the mean of the capability distribution, the association between capability and resources ($\rho = 0$), the tail of the resource distribution ($\alpha_R = 2$), the size of the applicant pool ($n = 50$), the noise of the resource signal ($\tau_R = 1$), and the form of the review signal (capability plus additive Gaussian noise, observed once). This appendix varies each of these and asks two questions: whether the ordering of fields by the value of targeting and of review survives, and what single quantity, if any, organizes the results. Every run uses two rounds, the default budget $b = 1$, the myopic funders with and without review, the complete-information benchmark, and $M = 1,000$ particles; 50 simulated populations per setting, 100 in the noise sweeps. The value of targeting is the complete-information optimum’s gain over uniform funding as a share of uniform funding’s gain over

TABLE S5. The gain from planning ahead and the forward-looking funder’s timing, by production function. Five rounds. The gain from planning ahead is the forward-looking funder’s gain over the myopic funder, both with review, as a percent of the research output that the forward-looking funder’s funding adds; the center of mass of spending is 0.5 when spending is even and larger when spending shifts toward later rounds. One hundred simulated populations per point; standard errors of the gain are at most 0.01 (Cobb-Douglas) and 0.4 ($\gamma = -3$) percentage points.

compounding rate ϵ	0.05	0.15	0.30	0.55	0.85
<i>Value of timing (percent of the gain from funding)</i>					
Cobb-Douglas ($\gamma = 0$)	-0.01	-0.01	0.00	0.01	0.03
$\gamma = -3$	0.05	0.56	1.96	4.56	6.52
<i>Center of mass of optimal spending</i>					
Cobb-Douglas ($\gamma = 0$)	0.499	0.500	0.501	0.505	0.512
$\gamma = -3$	0.511	0.540	0.577	0.623	0.660

TABLE S6. The value of review by field at a fixed Pareto scale and at a fixed mean capability. Share of the records-only shortfall recovered, at review noise $\tau_K = 0.3, 1$, and 3 ; 100 simulated populations per entry; $M = 1,000$ particles. The mean capability shown is the realized mean over the populations.

α_K	mean K	fixed scale 1			fixed mean 2			
		$\tau_K = 0.3$	1	3	mean K	$\tau_K = 0.3$	1	3
1.3	3.84	0.96	0.90	0.70	1.77	0.94	0.79	0.50
2	1.97	0.86	0.65	0.30	1.97	0.86	0.65	0.30
3.5	1.39	0.57	0.23	0.03	1.99	0.63	0.32	0.06

no funding; the value of review is the share of the records-only shortfall that the review signal recovers, as in Fig. 2C.

Mean capability. A Pareto distribution with a fixed scale has mean $\alpha/(\alpha - 1)$, so at scale 1 the heavy-tailed field ($\alpha_K = 1.3$) has mean capability 4.3, the default field ($\alpha_K = 2$) 2, and the evenly spread field ($\alpha_K = 3.5$) 1.4: a sweep of α_K at fixed scale varies mean capability along with inequality, and a higher mean makes resources more binding on its own. Table S6 compares that sweep with one in which the scale is set to $2(\alpha_K - 1)/\alpha_K$, so that every field has mean capability 2 (the realized means in the table fall short of 2 in the heavy-tailed field because the distribution has infinite variance; SI Materials and Methods). Holding the mean fixed narrows the heavy-tailed field’s advantage but preserves the ordering at every noise level; Fig. 2C and every number in the main text use the fixed-mean sweep.

TABLE S7. The value of review by field at equal informativeness. Share of the records-only shortfall recovered when the rank correlation between the review score and capability is held at the stated value; fixed mean capability; interpolated within the noise sweep of Table S6.

α_K	rank correlation	0.8	0.55	0.3
1.3	0.94	0.80	0.50	
2	0.85	0.65	0.32	
3.5	0.66	0.46	0.18	

Review compared at equal informativeness. The same review noise τ_K is a more informative signal where capability is more dispersed, because the noise then moves fewer researchers past one another. Table S7 therefore compares fields at equal informativeness, measured by the rank correlation between the review score and capability, interpolating within the fixed-mean sweep. At a rank correlation of 0.8, 0.55, and 0.3, the heavy-tailed field recovers 0.94, 0.80, and 0.50 of the shortfall, the default field 0.85, 0.65, and 0.32, and the evenly spread field 0.66, 0.46, and 0.18. The ordering is unchanged; the gaps between fields are smaller than at equal noise.

The association between capability and resources, the resource tail, and pool size. Table S8 varies the three remaining population parameters one at a time, at the default review noise and at Pareto scale 1. The association between capability and resources is the one that matters. As the copula parameter ρ rises from -0.5 to 0.8 , the value of targeting falls in every field (from 0.96 to 0.69 in the heavy-tailed field, from 0.25 to 0.06 in the evenly spread one), and so does the value of review; the ordering of fields is unchanged at every ρ . The mechanism is the gap rule itself: when resources already sit with the capable, the gaps $cK_i - R_{i0}$ are small and even, and there is little for targeting, or for the information that targeting requires, to add. The tail of the resource distribution changes the value of targeting by at most a tenth and the value of review by less. Pool size raises the value of targeting in the heavy-tailed field, because a larger pool contains more of the rare researchers of very high capability, and raises the value of review in the evenly spread field, where a larger pool gives review more to separate; neither effect changes the ordering.

The resource signal. The main text holds the resource signal at noise $\tau_R = 1$. Table S9 varies it from nearly exact ($\tau_R = 0.1$) to nearly uninformative ($\tau_R = 10$) and removes it altogether. Resource information matters least where review matters most. In the heavy-tailed field the records-only shortfall moves from 0.44 to 0.48 of the complete-information gain across the whole range, and the value

TABLE S8. The value of targeting (upper block) and of review (lower block) by field as the copula parameter ρ coupling capability and resources, the tail of the resource distribution, and the size of the applicant pool vary. Review noise $\tau_K = 1$; Pareto scale 1; 50 simulated populations per entry; $M = 1,000$ particles. Default values: $\rho = 0$, $\alpha_R = 2$, $n = 50$.

α_K	copula parameter ρ				resource tail α_R			pool size n			
	-0.5	0	0.5	0.8	1.3	2	3.5	25	50	100	200
<i>Value of targeting (share of uniform funding's gain)</i>											
1.3	0.96	0.91	0.82	0.69	0.92	0.91	0.82	0.81	0.91	1.06	1.15
2	0.56	0.49	0.38	0.25	0.43	0.49	0.46	0.44	0.49	0.53	0.57
3.5	0.25	0.20	0.13	0.06	0.17	0.20	0.18	0.20	0.20	0.21	0.22
<i>Value of review (share of the records-only shortfall recovered)</i>											
1.3	0.92	0.90	0.88	0.85	0.88	0.90	0.92	0.87	0.90	0.94	0.95
2	0.67	0.65	0.62	0.56	0.65	0.65	0.65	0.60	0.65	0.71	0.74
3.5	0.24	0.22	0.16	0.01	0.24	0.22	0.21	0.14	0.22	0.27	0.32

TABLE S9. The resource signal. The records-only shortfall (upper block) and the value of review (lower block) as the noise of the resource signal τ_R varies from nearly exact to nearly uninformative, and with no resource signal; review noise $\tau_K = 1$; fixed mean capability; 50 simulated populations per entry; $M = 1,000$ particles.

α_K	resource-signal noise τ_R					none
	0.1	0.3	1	3	10	
<i>Records-only shortfall (share of the complete-information gain)</i>						
1.3	0.45	0.44	0.45	0.47	0.48	0.48
2	0.25	0.25	0.27	0.29	0.30	0.30
3.5	0.09	0.09	0.11	0.14	0.15	0.15
<i>Value of review (share of the records-only shortfall recovered)</i>						
1.3	0.79	0.79	0.79	0.77	0.77	0.76
2	0.68	0.67	0.65	0.61	0.59	0.59
3.5	0.39	0.37	0.31	0.26	0.24	0.24

of review from 0.79 to 0.76: the researchers who matter there are identified by capability, and resources add little to their identification. In the evenly spread field a nearly exact resource signal cuts the records-only shortfall from 0.15 to 0.09, because once resources are known, output identifies capability; there the resource signal does a part of what review does, and the value of review falls from 0.39 with an exact resource signal to 0.24 with none. The ordering of fields is unchanged at every τ_R .

Multiplicative review noise. The main text's review noise is additive, so a given τ_K is a smaller relative error for a researcher of high capability than for one

TABLE S10. Multiplicative review noise. The review score is $K_i \exp(e_i)$ with e_i Gaussian of standard deviation s , and the funder knows the form; fixed mean capability; 50 simulated populations per entry; $M = 1,000$ particles. The upper block gives the informativeness of the score at each s .

α_K	noise s				
	0.1	0.3	0.6	1	1.5
<i>Rank correlation between score and capability</i>					
1.3	0.96	0.83	0.65	0.48	0.35
2	0.93	0.73	0.51	0.34	0.24
3.5	0.85	0.55	0.33	0.21	0.14
<i>Value of review (share of the records-only shortfall recovered)</i>					
1.3	0.96	0.88	0.67	0.43	0.27
2	0.89	0.71	0.42	0.21	0.09
3.5	0.68	0.39	0.15	0.05	0.02

of low capability, and the question is whether that form favors the heavy-tailed field. Table S10 replaces the score by $K_i \exp(e_i)$ with e_i Gaussian of standard deviation s , a constant relative error, with the funder knowing the form. The form does matter, in the direction the question anticipates. At equal informativeness, multiplicative noise costs the heavy-tailed field most: at a rank correlation of 0.65 between score and capability, review recovers 0.66 of the shortfall under multiplicative noise against 0.87 under additive noise in the heavy-tailed field, 0.61 against 0.75 in the default field, and 0.49 against 0.55 in the evenly spread field, because a constant relative error is a large absolute error for exactly the researchers who matter most in a heavy-tailed field. The ordering of fields holds at every s and at every matched informativeness, and so does the main text’s statement that noisy review retains most of its value under heavy tails: at $s = 0.6$, review retains 69 percent of its maximum value in the heavy-tailed field, 47 percent in the default field, and 22 percent in the evenly spread field. The heavy-tailed field’s advantage is smaller under multiplicative noise than the main text’s additive figures show, and it does not disappear.

Review observed once against review redrawn every round. In the main text each researcher’s review score is observed once and reused in every round, so a reviewer’s misjudgment persists (SI Materials and Methods). Table S11 redraws the score every round with independent noise, so that repeated review accumulates information the way the record does. Over two rounds the second review adds 0.01 to 0.04 to the share of the shortfall recovered, most where review is noisiest, and the ordering of fields is unchanged. Over twenty rounds the difference is larger: in the default field the review funder’s shortfall falls from 8 percent of the

TABLE S11. Review observed once against review redrawn every round. Share of the records-only shortfall recovered over two rounds at review noise $\tau_K = 0.3, 1$, and 3; fixed mean capability; 100 (once) and 50 (redrawn) simulated populations per entry; $M = 1,000$ particles.

α_K	observed once (main text)			redrawn every round		
	$\tau_K = 0.3$	1	3	0.3	1	3
1.3	0.94	0.79	0.50	0.95	0.82	0.52
2	0.86	0.65	0.30	0.88	0.68	0.32
3.5	0.63	0.32	0.06	0.66	0.36	0.07

TABLE S12. The value of review when the review score partly reflects resources. Share of the records-only shortfall recovered when the score is $K_i + \beta R_{i0}$ plus noise, for $\beta = 0$ (the main text’s signal), 0.25, 0.5, and 1; fixed mean capability; 50 simulated populations per entry (100 at $\beta = 0$); $M = 1,000$ particles.

α_K	$\beta = 0$	$\tau_K = 0.3$				$\beta = 0$	$\tau_K = 1$		
		0.25	0.5	1			0.25	0.5	1
1.3	0.94	0.92	0.88	0.76	0.79	0.77	0.75	0.70	
2	0.86	0.82	0.75	0.55	0.65	0.62	0.59	0.52	
3.5	0.63	0.55	0.42	0.26	0.32	0.27	0.23	0.17	

complete-information gain in round one to 3 percent by round twenty when the score is observed once (Fig. 2B), and to 0.8 percent when it is redrawn every round; in the heavy-tailed field, to 2 percent and to 0.7 percent. The records-only funder’s shortfall in the same runs falls from 46 to 19 percent (default) and from 60 to 14 percent (heavy-tailed) either way. Repeated independent review would therefore strengthen the main text’s comparison, and the once-observed score is the conservative case.

Review that partly reflects resources. The main text’s review signal observes capability with noise. Real review may also reflect the resources a researcher visibly has. Table S12 replaces the signal by $K_i + \beta R_{i0} + \xi_i$, with the funder knowing β and updating accordingly. Contamination costs least where capability is heavy-tailed: at $\beta = 1$, where the score weights resources as heavily as capability, the heavy-tailed field still recovers 0.70 to 0.76 of the shortfall, against 0.79 to 0.94 for a clean signal. It costs most where capability is evenly spread, where at $\beta = 1$ review recovers 0.17 to 0.26 of the shortfall, against 0.32 to 0.63 for a clean signal. The reason is the one behind Fig. 2C: the researchers whom review must find in a heavy-tailed field stand so far apart in capability that a resource term does not hide them.

The inequality of the gaps organizes the results. The runs above vary the population parameters in five ways. One quantity summarizes what they do to the value of targeting: how unequally the optimal grants, the gaps $\max(cK_i - R_{i0}, 0)$ at the population's budget, are distributed across the field. Across all 159 population settings in this appendix (mean capability, association, resource tail, pool size, and the two noise sweeps), the logarithm of the value of targeting is a nearly linear function of the Gini coefficient of the optimal grants, with $R^2 = 0.96$ (Fig. S7); adding the ratio of mean capability to mean resources raises R^2 to 0.99. Settings that differ in the capability tail, the association between capability and resources, the resource tail, and pool size, but agree on the inequality of the gaps, have the same value of targeting. The value of review depends on the same quantity together with the signal's informativeness: the log-odds of the share recovered, regressed on the Gini coefficient of the optimal grants and the log-odds of the rank correlation between score and capability, has $R^2 = 0.90$, and adding the capability tail as a separate variable raises it by less than 0.01. Capability inequality is the main driver of gap inequality in every sweep we ran, which is why the main text can speak of it directly; the association between capability and resources is the second driver, and it works through the same channel. Two cautions apply to using this quantity. The gaps depend on the budget, through c , as well as on the population, so the inequality of the gaps is a property of a field and a funder together; and it is a latent quantity, computed from capabilities and resources the funder does not observe, so it organizes the model rather than serving as a statistic a funder can estimate directly. The main text's Discussion lists the observable quantities that stand in for it.

SI Text 6. STRATEGIC TIMING OF FUNDING

The funder holds a fixed budget for the whole horizon and may time its spending freely. The strongest argument for spending early concerns fields whose researchers have few resources: a researcher without resources produces nothing, so the funder observes nothing and no capability grows, and money held for later rounds produces neither evidence nor growth in the meantime. The argument fails. A funder that spends everything in the first round allocates before observing any research output, and knows exactly as little as a funder that spends everything in the last round; in a field with no resources at all, the two produce identical expected output. What the argument does establish is a complementarity: early spending enables observation and growth, and late spending exploits what early spending revealed. That favors putting some money early. It does not favor putting most of it early.

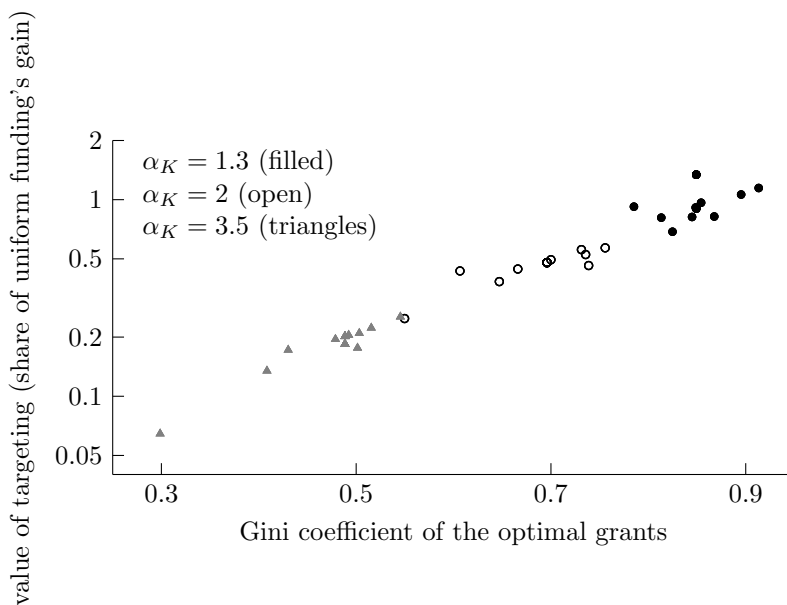


FIGURE S7. The inequality of the gaps organizes the value of targeting. Each point is one setting of SI Text 5: the value of targeting, on a logarithmic scale, against the Gini coefficient of the optimal grants at that setting's budget. The 159 settings vary the capability tail (marker), the correlation between capability and resources, the tail of the resource distribution, the size of the applicant pool, and the scale of the capability distribution; the logarithm of the value of targeting is linear in the Gini coefficient with $R^2 = 0.96$.

In our simulations, under complementarity, observation comes first and money second. Wherever capabilities compound at more than a negligible rate (ϵ above about 0.03), the forward-looking funder spends less than an even share in the first round and spends the bulk once output has revealed who is capable; scarce baseline resources raise the early share and weaken the effect, but do not reverse it (Fig. S8A). The shift of spending toward later rounds comes from the output researchers produce whether or not they are funded, which compounds and reveals capability while the funder waits (SI Text 3, Fig. S4).

Planning ahead is worth less than the signal provided by grant peer review, in every setting we test. Across all horizons and compounding rates in our sweeps, the forward-looking funder's gain over the myopic funder, which spends equal installments each round, stays below the review signal's value: by an order of magnitude or more at the default compounding rate and below, and approaching the signal's value only at the highest compounding rate and the longest horizon we consider (Fig. S8B). As a share of the research output that the forward-looking funder's funding adds, the gain from planning ahead ranges from indistinguishable

from zero to 4.5% across the 30 settings tested; the review signal’s value in the same settings ranges from 7% to 27%. The gain exists only where capability and resources are complements: under Cobb-Douglas production it is indistinguishable from zero (SI Text 4, Table S5).

Three limits bound these results, and they are the reason the timing analysis sits in the SI rather than the main text. First, the forward-looking funder is a certainty-equivalent planner (SI Materials and Methods): it treats future output as equal to its expectation and does not value the information its own grants will generate, so it is not the Bayes-optimal sequential policy, and its gain is a lower bound on what planning ahead could be worth. Second, the gain combines two changes at once, when the money is spent and whom it reaches, because a funder that anticipates capability growth also values recipients differently; we do not separate the two, and “gain from planning ahead” names their sum. Third, the objective counts output within the horizon only and assigns no value to the capability that remains at its end. A terminal value for capability would credit the capability that late grants build, which the horizon objective ignores, and also the compounded capability that early grants leave behind; we have not measured which effect is larger, so the size of the shift toward later rounds should be read as specific to the horizon objective.

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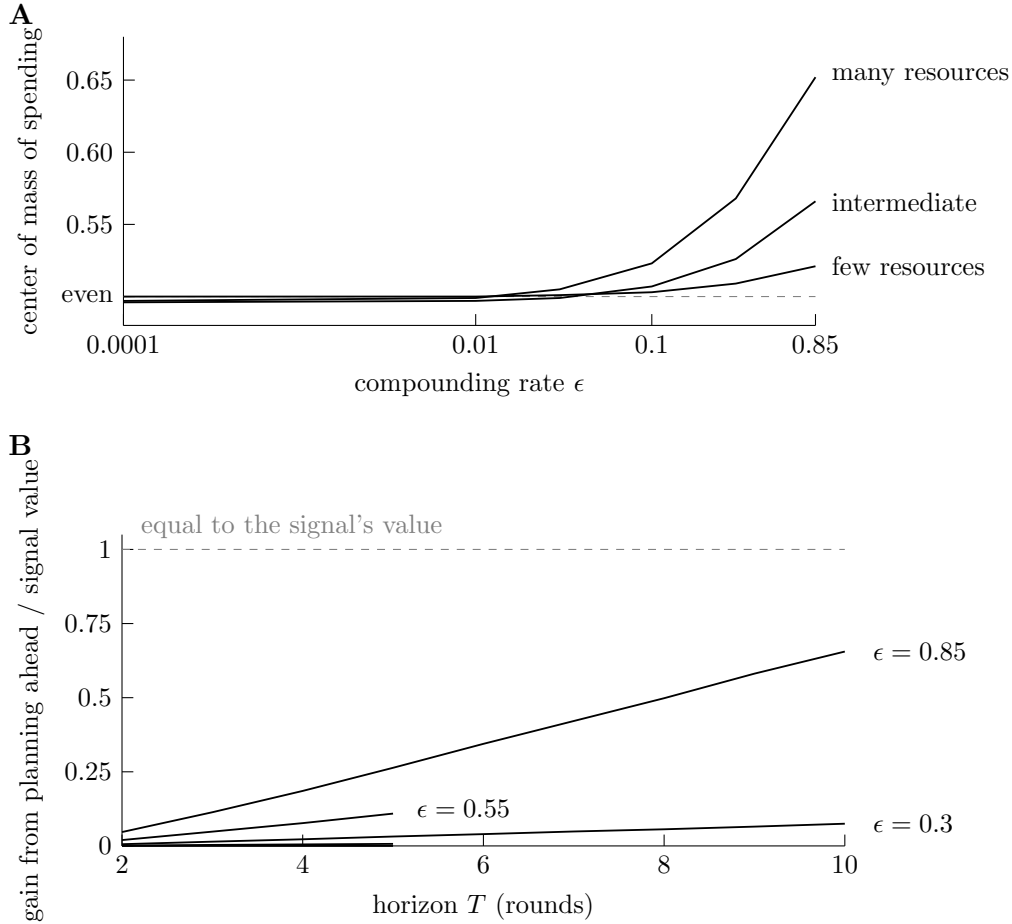


FIGURE S8. Strategic timing of funding. (A) The compounding rate, not the scarcity of baseline resources, sets how far spending shifts toward later rounds: the center of mass of spending (0.5 is even; higher is later) as the compounding rate ϵ increases, for three baseline resource scales. (B) Choosing whom outweighs choosing when: the gain from planning ahead, as a share of the review signal's value, by horizon and compounding rate; in every setting the gain is below the signal's value. Settings: SI Materials and Methods.

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